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Buildings illustrated

CI/SfB 054

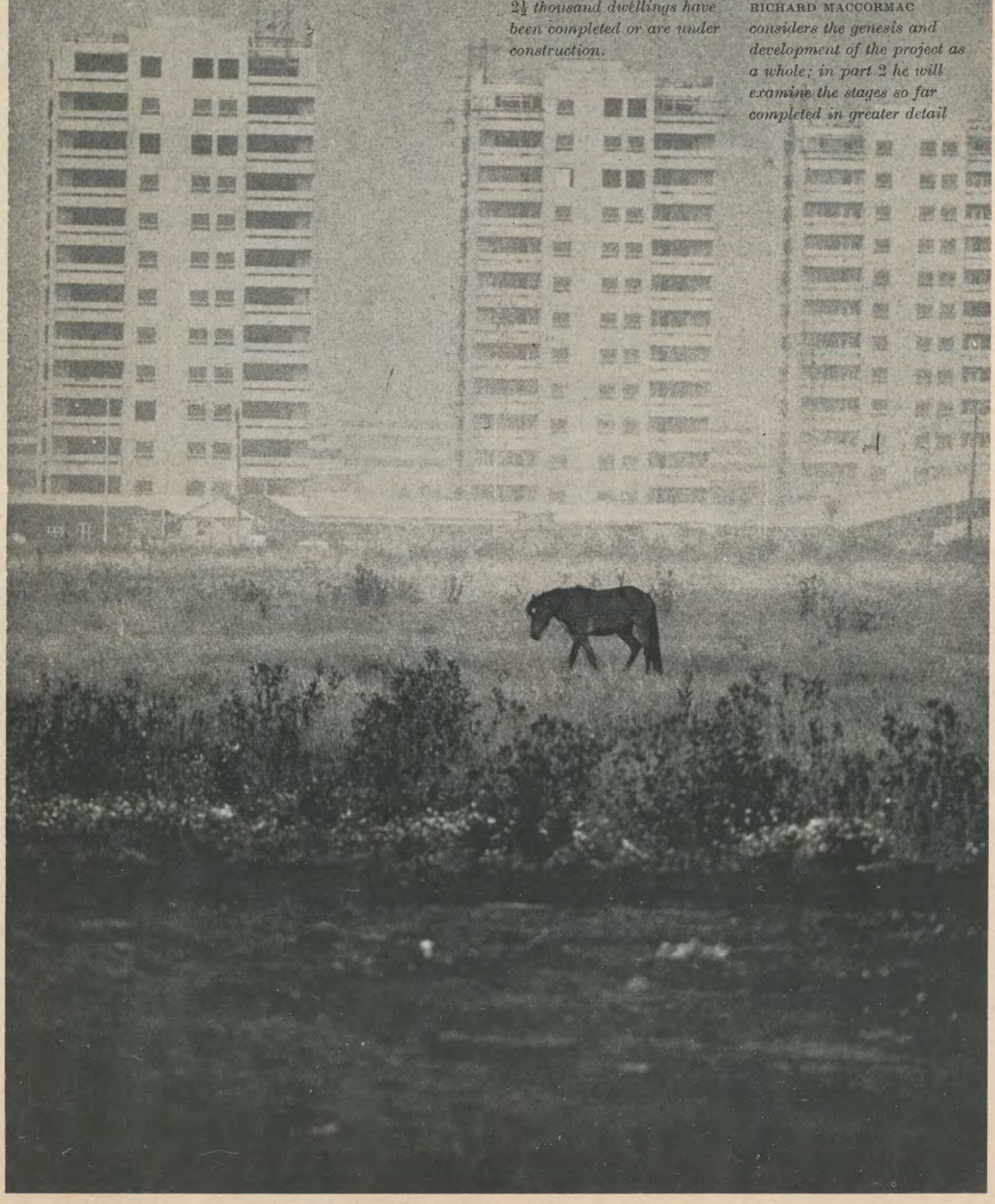
New town

GLC Department of Architecture and Civic Design

Thamesmead: part one

Seven years ago the initial master plan for Thamesmead proposed what was in effect a self-contained new town within Greater London for 60 000 people. Already over 2½ thousand dwellings have been completed or are under construction.

Thamesmead is very much a phenomenon of the 1960s (indeed, if conceived today, it must be doubted whether so ambitious a project could clear administrative hurdles). This week RICHARD MACCORMAC *considers the genesis and development of the project as a whole; in part 2 he will examine the stages so far completed in greater detail*

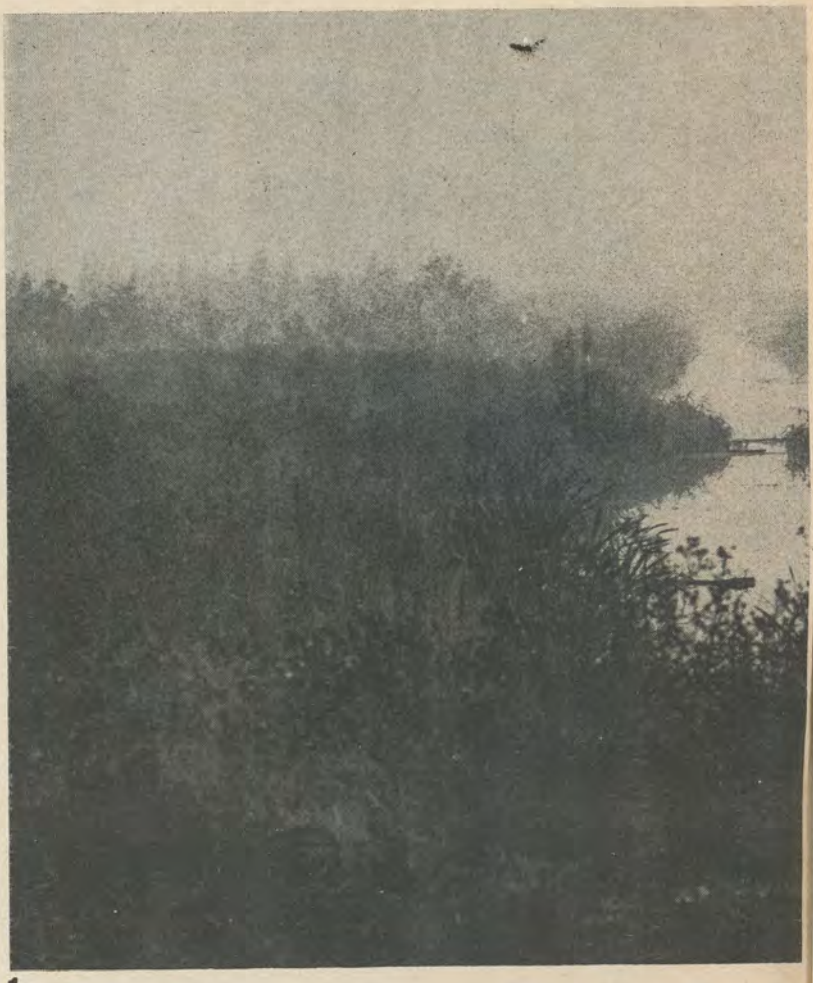


THAMESMEAD—A RIVERSIDE DEVELOPMENT

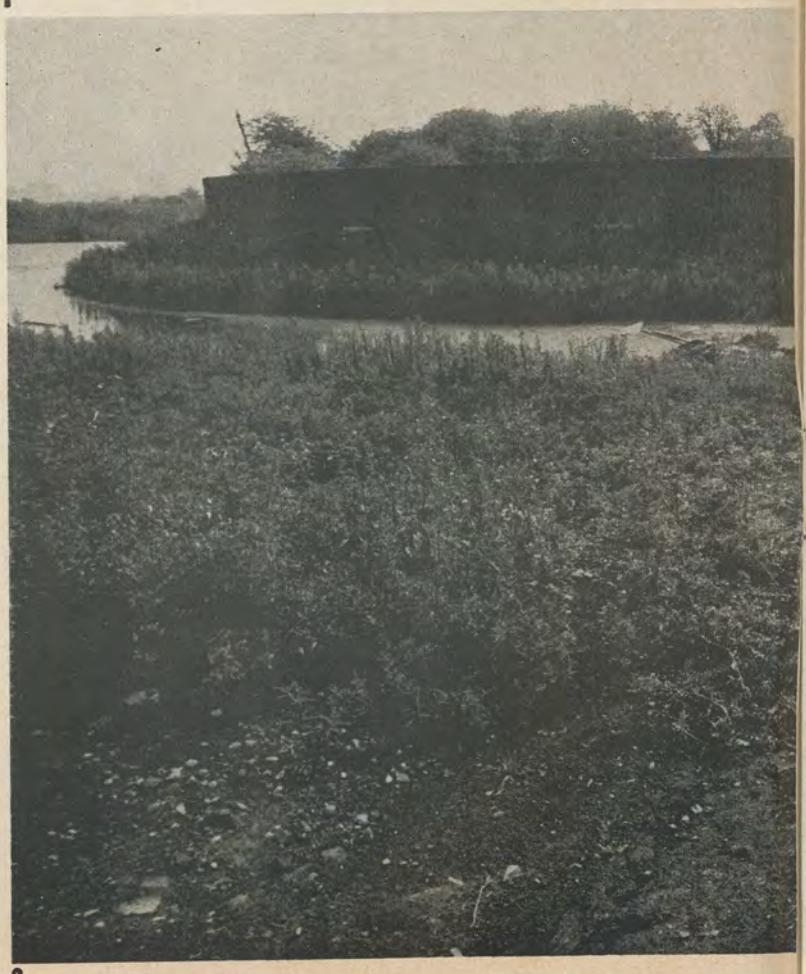
For the Greater London Council
This has been a team effort in every sense of the word but the following officers had a major responsibility in the project:
architect to the council Sir Roger Walters in succession to Sir Hubert Bennett
traffic commissioner and director of transportation P. F. Stott
director of planning B. J. Collins
chief officer of parks department J. C. Kennedy in succession to the late F. Hallowes and W. Wilkes
director of M & E services C. A. Belcher
valuer and estates surveyor Kenneth H. Blessley
public health engineer S. H. Dainty
Department of Architecture and Civic Design:
deputy architect Jack Whittle
Thamesmead manager Geoffrey Horsfall
Thamesmead architect J. G. H. D. Cairns
assistant divisional architect D. T. Grove
principal housing architect Kenneth J. Campbell
production team: P. A. Bottomley, N. Engleback, John Knight, R. E. Rigg, J. A. Roberts, G. A. Comrie-Smith, P. A. Westwood
quantity surveyor H. S. Page in succession to M. F. Rice
housing engineer J. R. Stanley

ARCHITECTS' ACCOUNT—DEVELOPMENT OF THAMESMEAD MASTER PLAN

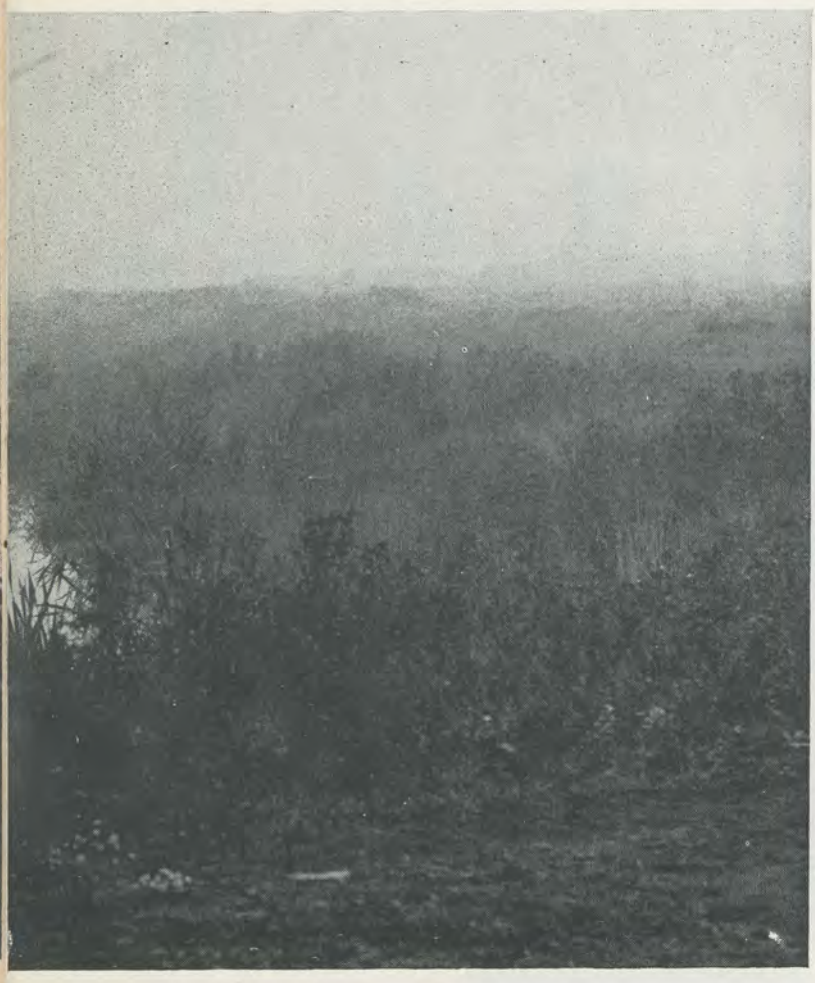
Towards the latter part of the 1950s the major comprehensive redevelopment areas created primarily to rebuild the war-damaged East End were nearing completion and the LCC was searching for further sites where large scale housing gain could continue to be made. All surplus or undeveloped land within London was investigated and some of the first to be looked at was already in council ownership.
The reserve land surrounding the Cross Ness sewage treatment works on the Erith Marshes had originally been required for extension of the works. However, their complete re-design and reconstruction between 1950 and 1960 left about half the reserve land available for development. The proximity of the sewage works was, unfortunately, not the only problem attached to this site. In 1953 it had been flooded when the river wall was breached downstream, and the borough of Bexley had imposed a by-law on all its low lying lands adjacent to the river prohibiting the building of any habitable rooms below possible flooding level. Apart from this, ground conditions were bad, as the whole site was marshland with standing water 4ft to 5ft below normal ground level and all was on a 20ft to 25ft bed of peat. Piling would be required for the buildings and for drainage. It was to a considerable extent treeless and featureless apart from the 20ft bank to the north **21** containing two 11ft diameter brick sewers heading for the Cross Ness works **3**. Communications with central London were poor. The North Kent line of British Rail was on the edge of the site to the south but it was already working almost to capacity and this was equally true of the road system. The nearest river crossing was Woolwich Ferry.
A scheme to house 25 000 in a mixture of 31-storey towers and clusters of lower blocks, built on podiums over car parking and servicing areas, was presented to committee in 1963. But it was soon shelved as at this time, following many years of discussion with the Ministry of Defence, the council was starting negotiations for the purchase of surplus land in the Woolwich Arsenal, adjacent and to the north of Erith Marshes. At first it seemed that only the

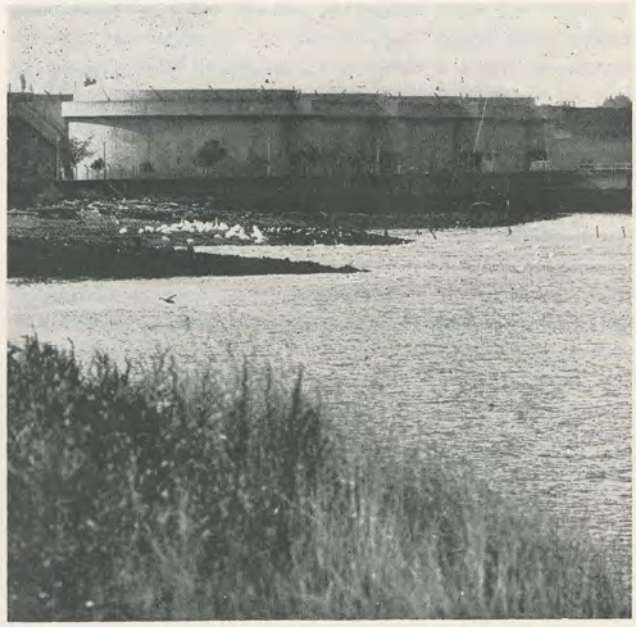


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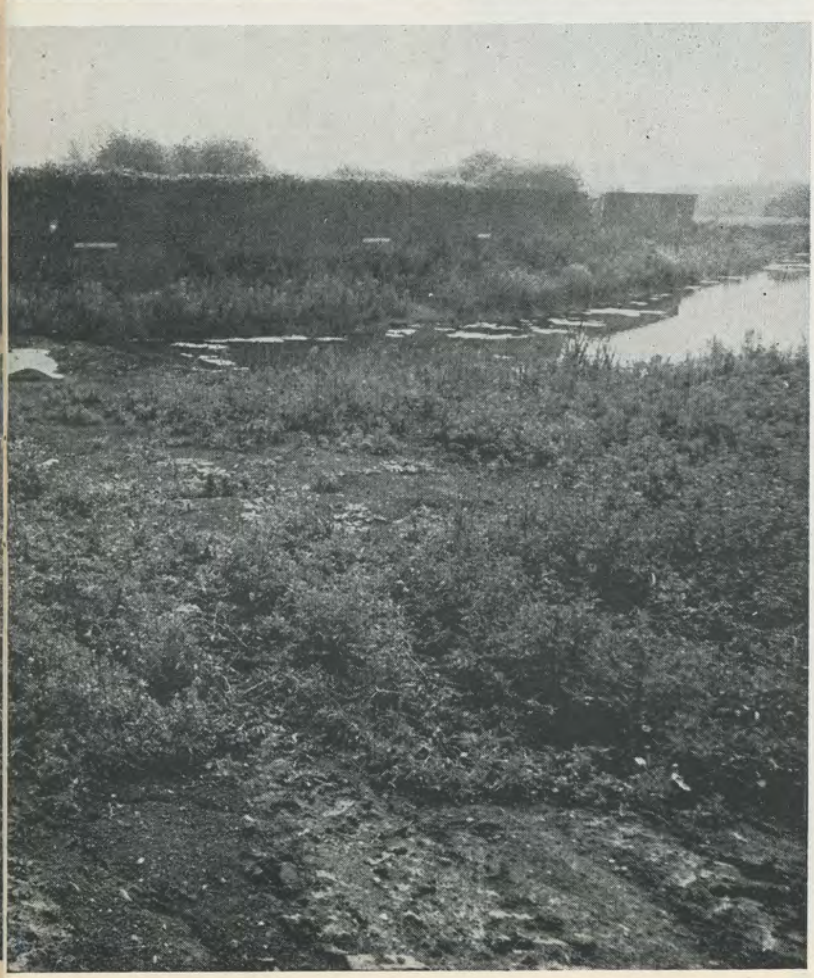




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eastern end of the Arsenal area was to be offered but gradually more sections were added until there were more than 1000 extra acres similar in quality to the Erith site, but with three miles of river frontage **9**. It was now apparent that the council was involved in a development of new town proportions and a team made up of staff from the Planning and Architect's Departments was set up to make an overall master plan.

Among their early decisions were that the density of the development, because of London's great housing shortage, should be up to an average 100 persons per acre overall housing land (30 ppa higher than surrounding zoning); that a job should be provided within the town for every household to ensure that as few people as possible were thrown on to the overloaded communications system into central London; that there should be a bus route within ¼ mile of every home; that there should be complete pedestrian-traffic segregation; that as many homes as possible should have views of the river; and that canals and lakes should be used as amenity features serving a dual role as a surface-water drainage system **1**, **2**.

1 *Thamesmead site: flooded marshland in middle of industrial area*
2 *Memento of former use: one of several disused ammunition dumps from Woolwich Arsenal, to be preserved as features in new plan*
3 *Cross Ness sewage treatment works which adjoins east boundary of town*
4 *Thames estuary and conurbation showing Thamesmead on direct line from central London to Foulness. (Primary road network based on* GLC*'s latest proposals)*

Provision for such things as schools, playing fields and public open space were based on national or regional standards, but the decision to create 10 local centres as well as the main town centre was based on research on social and community facilities carried out for the council's previous, but abandoned, new town at Hook in Hampshire. Much experience was brought into the new team by planners and architects who had worked on that scheme **16**.

Cumbernauld had a more indirect, but equally strong influence at this time, and the three-dimensional density-structured plan of Thamesmead **10** as well as the multi-level town centre **13** show a similar architecturally-based planning concept.

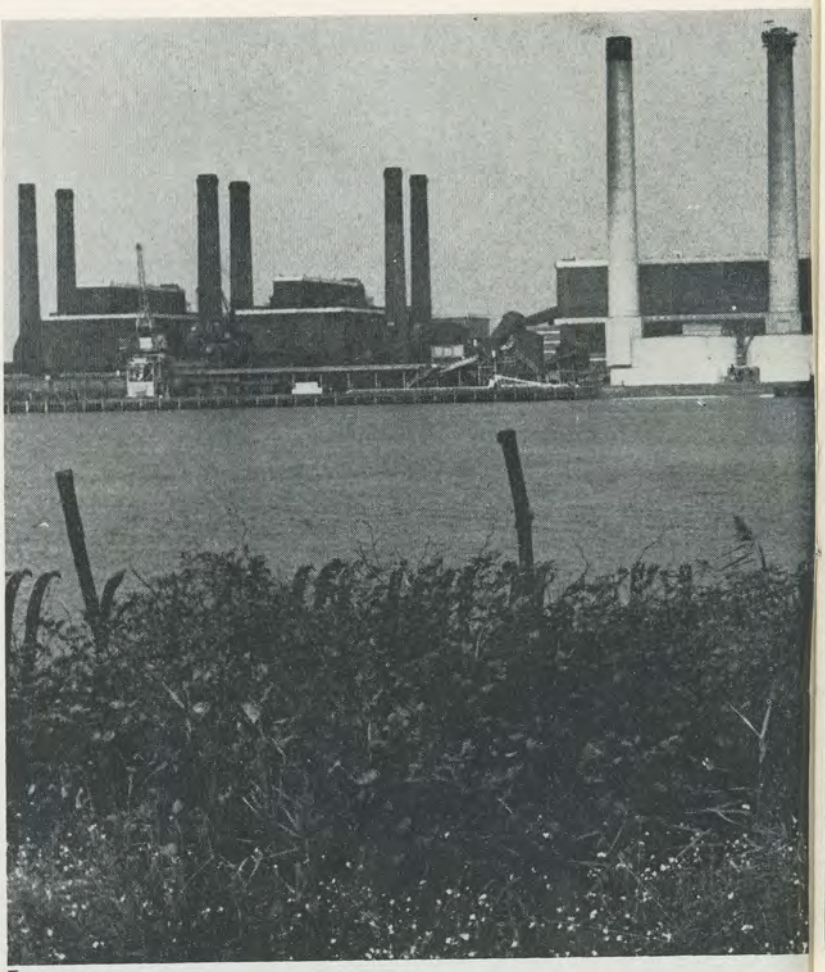
The linear quality of the town as shown in the town model **11** produced and presented at the same time as the final master plan, reflected the then current feeling for social and architectural integration and continuity shown in the Park Hill housing at Sheffield **17** and as a reaction from the isolation of the Corbusian type of development with tall blocks set in areas of open space. It also sited as many homes as possible near the river in blocks which form a barrier to the wind. The strong structure of the town will be apparent from the hills to the south **24** as well as from within the town itself, and will give identity and interest in a featureless landscape.



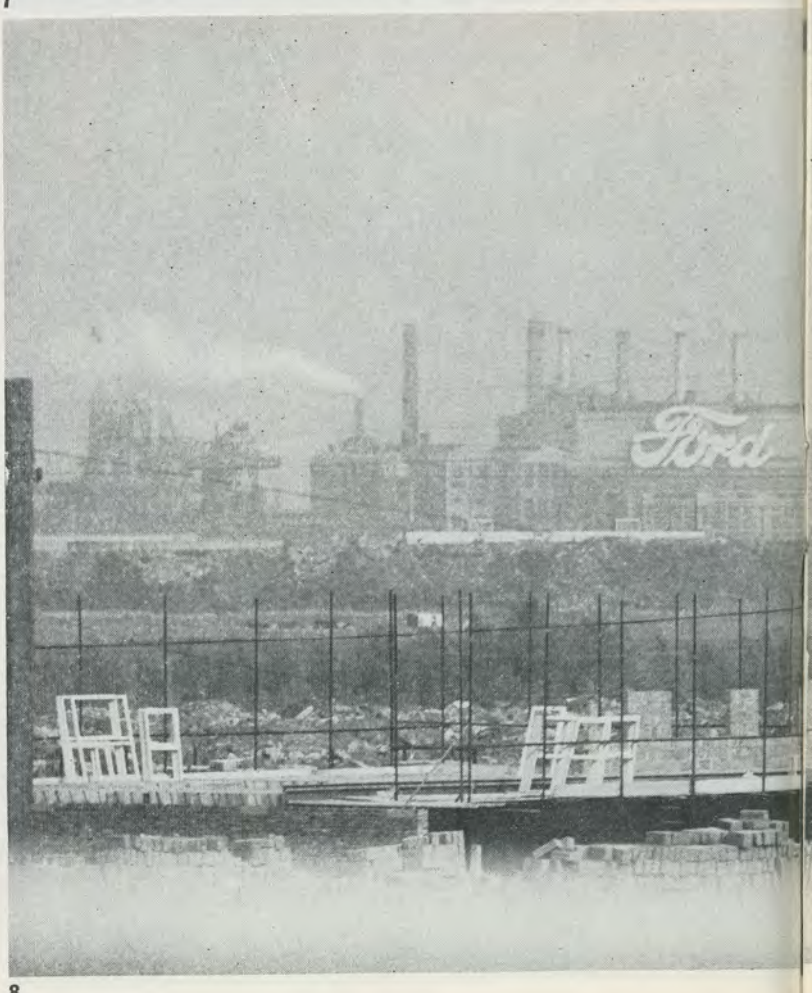
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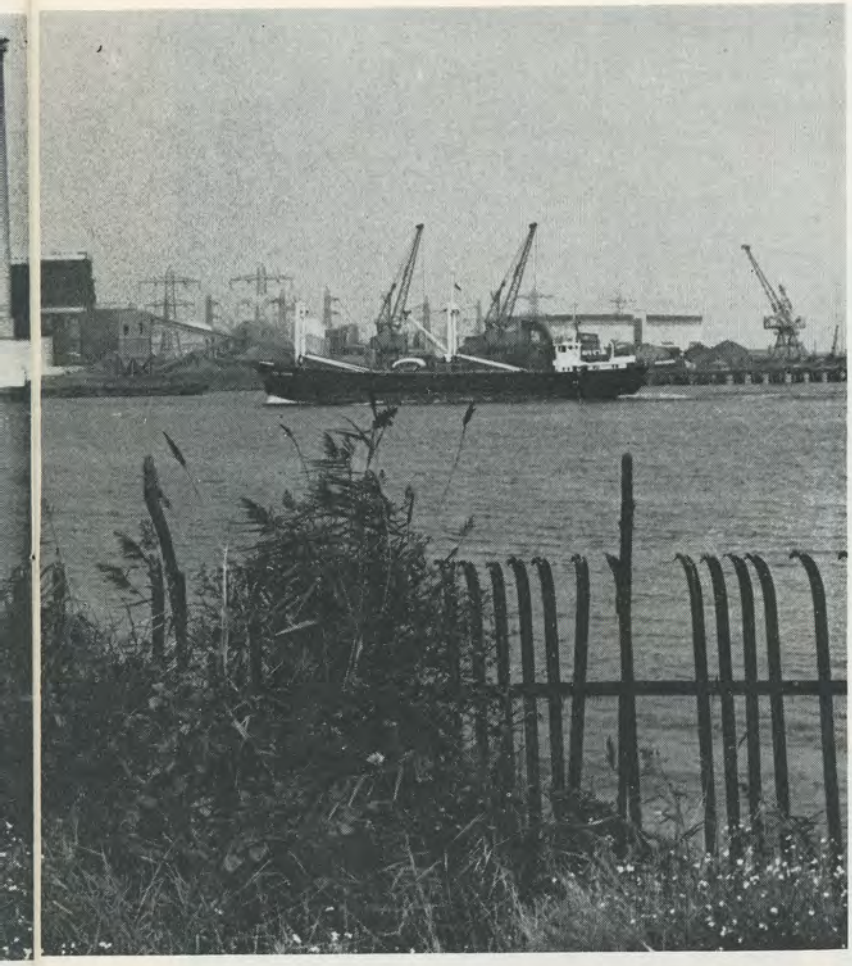
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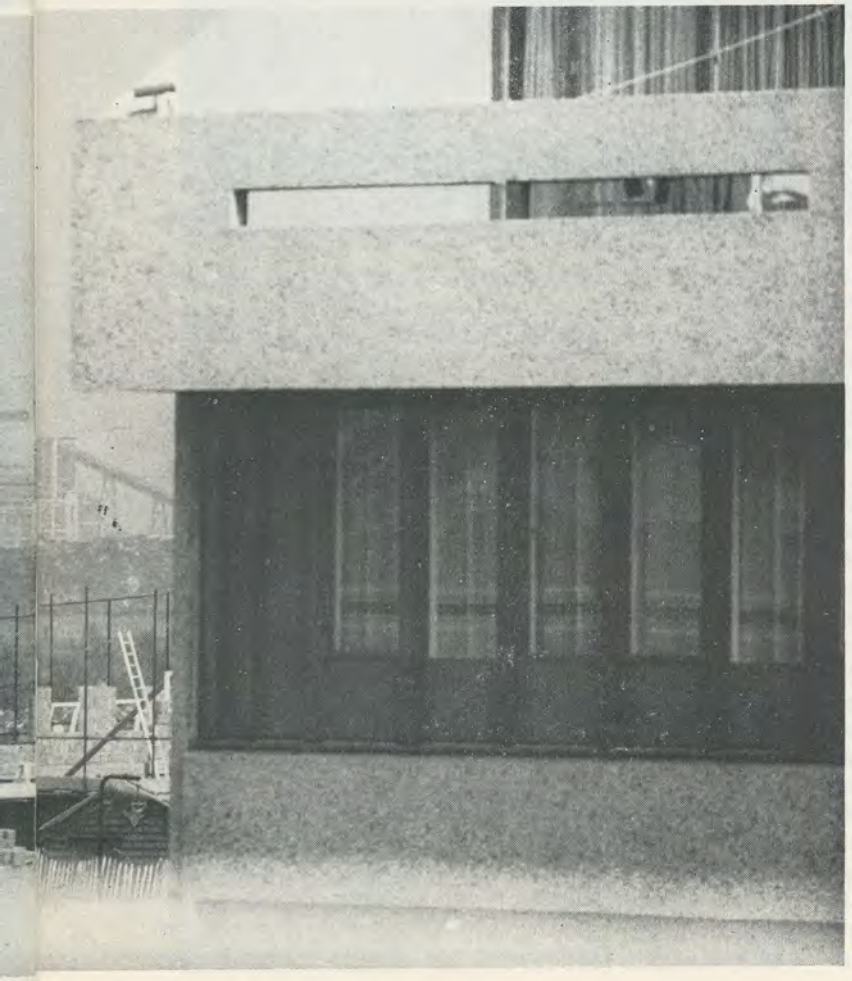


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APPRAISAL

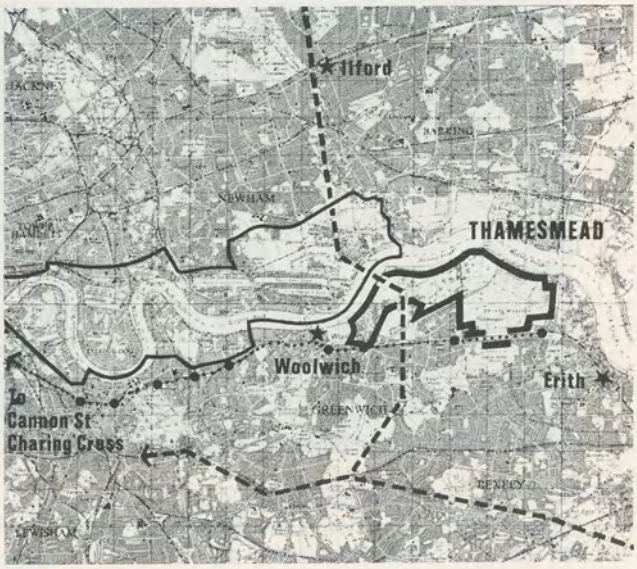
by Richard MacCormac

Superficially, it seems absurd to build a new community at Thamesmead, on 20ft of peat, next to a major sewage works **3** and under an umbrella of pollution from Barking and Belvedere power stations **7** (on the north bank), which is obnoxious enough to prohibit building above 200ft.

Yet, since 1965, when the surplus Royal Arsenal land was bought from the Ministry of Defence, the good sense of the GLC's decision has been confirmed. The choice of wasteland gives Thamesmead the topical virtue of resource conservation unlike its unrealised predecessor Hook, which would have swallowed several thousand acres of agricultural land. Instead of being an isolated community in an unwholesome industrial area, it now looks as though it will become part of a general redevelopment of the river east of the City on a line between London and Foulness **4**.

The inflation of the value of building land makes the original price of Thamesmead land seem incredible. At £170 per dwelling, the cost is low enough to offset the

The appraiser is an architect in private practice and lecturer at Cambridge University School of Architecture. He was joint winner of last year's Huddersfield Building Society Award for new ideas in housing

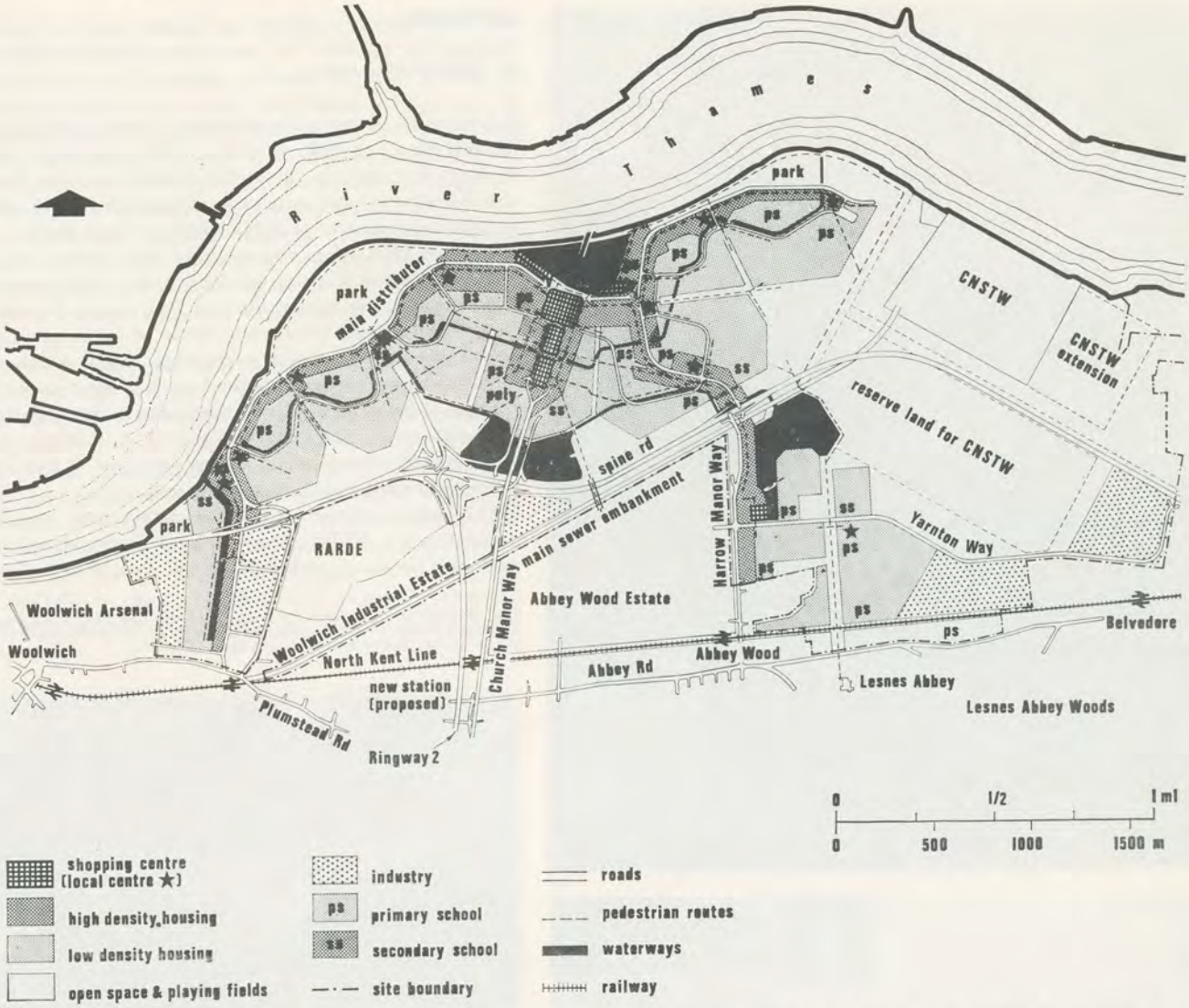


9

5 *Proposed tunnel under Thames will help relieve labour shortage on north bank. Here workers from Thamesmead area must rely on ferry,* **6**

7 *Industrial landscape dominates northern horizon across river: Belvedere and Barking power stations, and* **8** *Ford factory at Dagenham*

9 *Thamesmead site in relation to docklands study area, transport link and nearest strategic shopping centres*



10

exceptionally high development cost associated with piling buildings, roads and drains, so that the total costs are comparable with those of developing an inner London site.

The context of Thamesmead

Apart from its scale and the daunting technical problems of the site, the distinguishing objective of Thamesmead is that it should be a multi-class community with its own services, employment and amenities rather than a residential dormitory. It follows the social concepts of Hook and sets out to avoid the monolithic one-class society of the huge pre-war estates such as Dagenham.

Yet, unlike the new towns, Thamesmead is part of the existing matrix of south-east London **9**. It has to come to terms with the eight lane penetration of Ringway 2 and the six lanes east/west of the Woolwich/Erith road, both of which impose severe acoustic and visual constraints upon the site. Once the position of Ringway 2 was decided the outcome of the bridge versus tunnel argument became crucial to preserving the town's linear form along the river. The east/west road, on the other hand, was accepted on the Thamesmead site rather than further south (as originally planned) because the design team realised that it could be accommodated in a new situation with less injury. Even so, the basic form of the town is fundamentally, and I think adversely, affected by the status of the road.

These roads are also Thamesmead's life line. Since the closure of the Woolwich Arsenal in 1965 and the unexpected closure of the AEI factory in 1968, there has been a lack of skilled employment in the area and a feeling of industrial depression and isolation. Residents of Thamesmead travel far afield to work at present and, if both husband and wife work in central London, this will add an annual fare bill of more than £150 to the already high cost of shopping and rent.

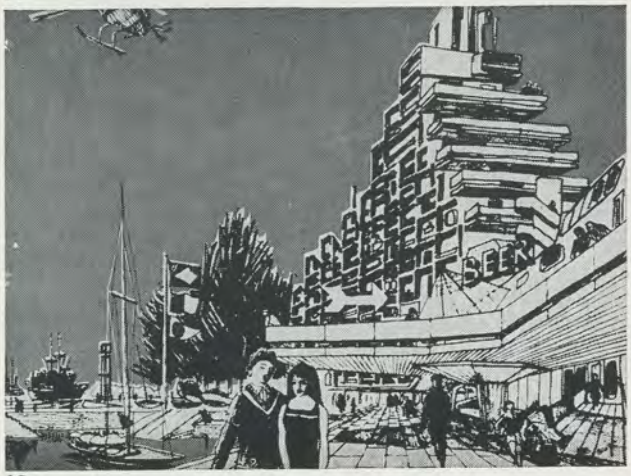
The recent curtailment of the ringway system **4** gives priority to Thamesmead's section and will connect the town with the area north of the river where there is currently labour scarcity **5, 6**. The east/west road should similarly open up a hinterland of employment. Conversely these improved connections should encourage the establishment of new industries and offices in the area. It is important that they do so, for British Rail's North Kent line, the only railway connection to central London, is already overcrowded.

Thamesmead's own shopping provision is complicated by the existence of the Woolwich and Erith centres **9**, a conflict with whose interests brought about a public inquiry in 1968. The outcome was to restrict the size of the new centre to 200 000 sq ft to be implemented in two phases of 100 000. Although the road system is designed to serve more than twice this area, Thamesmead is restricted, for the time being, to district rather than 'strategic' status.

There is another, different, sense in which Thamesmead is subject to its context. Although the first generation of residents have come from dense substandard housing, the general housing densities in the Woolwich/Erith areas do not go much above 50 ppa. To respond effectively to housing needs, Thamesmead was planned to double this



11



12



13

density. But ultimately the accommodation it offers will be judged in relation to the surrounding areas of pre-war speculative housing and the GLC's post-war Abbey Wood estate at 70 ppa. In terms of density this remains a predominantly suburban rather than urban area **22**.

The master plan

The identity of Thamesmead is therefore ambiguous. As a sudden administrative *tour de force* springing up out of marshland it is distinct; but to the residents who go to the cinema in Woolwich or work in Dagenham **8** it is probably only part of an extensive personal schema of locality.

The evolution of the master plan to some degree reflects this dilemma. The GLC planners' brief to the architects in May 1965 proposed a centre on the north/south axis of Harrow Manor Way, which would have linked with the railway station and a subcentre at Abbey Wood; one of the objectives was the integration of the existing Abbey Wood estate.

In the published report (*Woolwich-Erith: a riverside project* GLC) of March 1966 other criteria have prevailed: 'the need to seize the visual opportunity that the Thames water front offers' has become a priority. In the 1965 draft plan the half-mile wide band of housing along the river is not structured by density, whereas the 1966 published master plan proposes a high-density spine along the front 'giving the new town a firm riverside identity, and creating a scale of development to balance the structures on the north bank' **12**. The centre is also located along the river front at the densest part of the spine where shops and communal buildings become part of the river promenade and overlook a yacht station for the Port of London Authority.

One detects two changes of emphasis in the plan, both of which continue to determine the nature of the present development. It is responsive to the particular quality of the site, flatness, water and dramatic scale; and it has become more independent of its context having a completeness of form on the north side of the sewer bank which leaves the first stages and Abbey Wood as appendages **26**.

10 *Thamesmead master plan in its* 1967 *form*
11 *Model of master plan* (1967 *version*) *looking east shows dominance of spine along river frontage* (*park in foreground left required for shipping sightlines*). *Abbey Wood estate and stages* I *and* II *are beyond sewer-bank crossing diagonally in background*
12 *Vision of an urban landscape* (*from* 1966 *Woolwich-Erith report*). *Highest density concentrated along river*
13 *Part of model from south showing central area, marina and ringway* 2 *entering tunnel*

The subsequent plan **10** remains much the same except that the centre has reverted to an axial form **13** so that the spine should shelter it from the severe winds which blow up the river; and the system of one-way distributor roads looping round the environmental areas has been transformed into a district distributor along the north edge of town making half loops through the middle of the environmental areas with culs-de-sac off them. Pedestrian routes and local roads converge in 'crows feet' at the junctions between environmental areas to create areas of high activity, for the location of pubs, shops, offices and bus stops, in association with the high-density housing spine. In contrast primary schools are associated with low-density housing and amenity areas for recreation. Playing fields and public open space buffer the low-density fringe from the east/west road.

Clearly the most important concept in the Thamesmead plan is this density structure and the most critical, the spine building **18** which forms the skeleton of the town. The arguments in favour are elaborate and interlocking, and need to be scrutinised one by one.

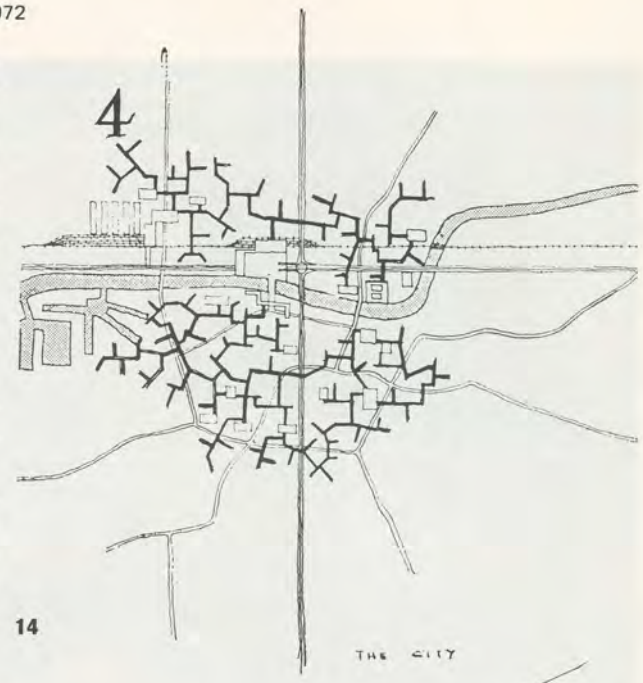
The visible form of the town

Already stages I and II dramatically illustrate the visual intentions of the plan. In the otherwise flat landscape, the spine buildings progress northward like trains towards the future centre of the town **28**. In Kevin Lynch's terms (in *Image of the city*) they are an artificial topography which identifies the places of highest activity, the vehicular and pedestrian routes to the centre and the environmental areas between them **19**. They enable the whole scale of the town to be understood from the open space at the south edge to the hard protective definition by the river. This is impressive and must overcome the kind of monotony which one experiences in large, dense, low-rise schemes such as Albertslund **29**. But a difficult question of values becomes apparent. At what cost and what sacrifice to residential amenity has a grand visual idea prevailed? Would it have been possible, with a low-rise scheme, to have achieved a structure through changes in the scale of open space, like the 18th century use of parks, squares, streets and mews? The architects think not, principally because their open space resource is largely allocated to noise protection. They also feel that the high buildings are needed for the town to hold its own against its neighbours, the power stations. This is true in a grand conceptual sense, but perceptually a row of apple trees 30 feet from a ground floor window could obscure the horizon.

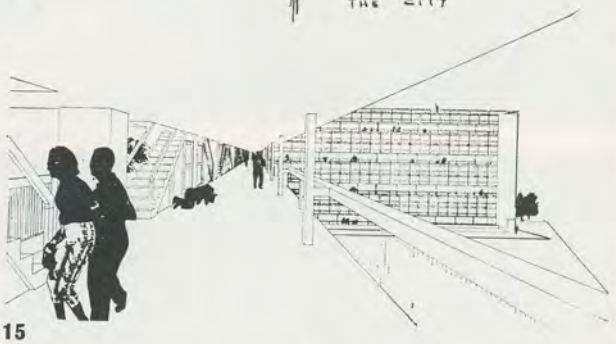
Spine buildings and the road system

The district distributor road along the north edge of the site will carry between 500 and 2000 vehicles an hour and the spine buildings are intended to protect the environmental areas from traffic noise **20**. Although the chosen hierarchy of roads may be an inherently good one it is also clearly the result of the density structure preferred for visual reasons. Without this structure the town could have been served by a more even mesh of less loaded and noisy distributors. One objection would be more frequent severance of environmental areas. But in fact, although the development plan seems to resemble a large Radburn diagram, the local distributors do penetrate residential areas. If the highway engineers have their way and apply their standards, the severance will be considerable in any case.

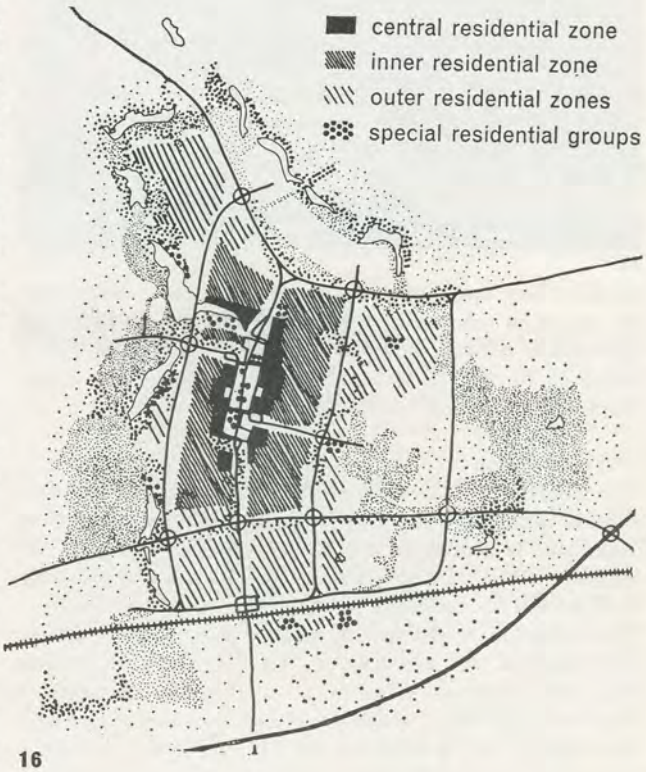
The use of medium-rise buildings as high-density acoustic barriers is also inherently difficult. To be effective they need to be single aspect; the greater frontage halves their density and increases cost. In stages I and II



14



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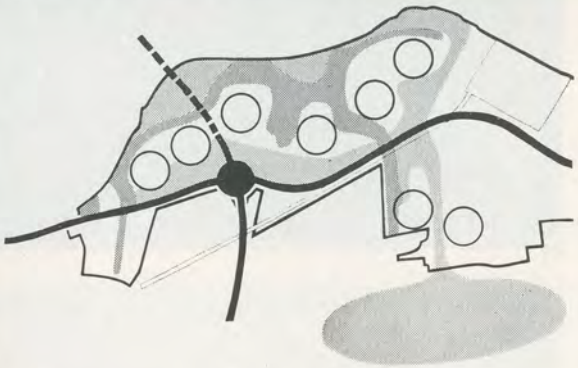
14 *Peter Smithson's first cluster city diagram (from* Team 10 primer *by permission of* Architectural Design)
15 *A & P Smithson's drawing of deck housing from* Team 10 Primer
16 *Diagrammatic plan of Hook new town* (1961) *shows build-up of density towards linear centre*



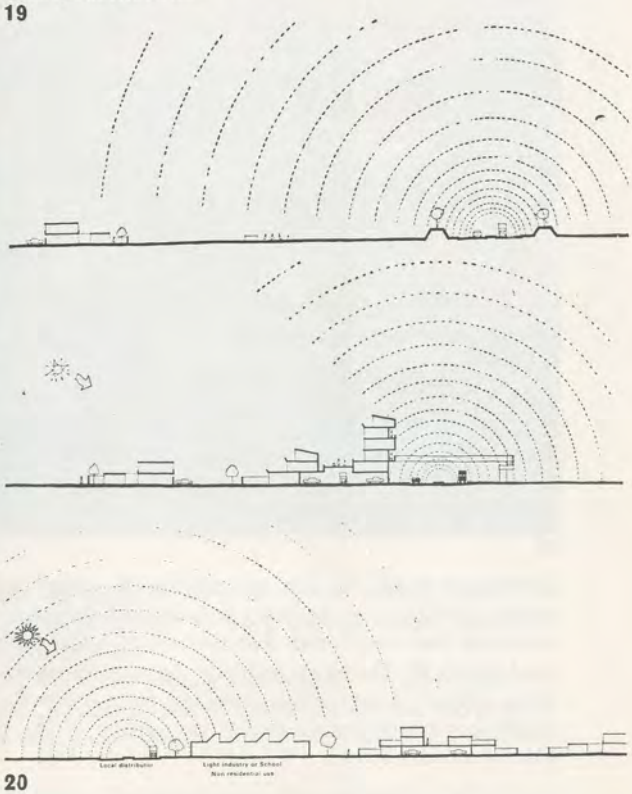
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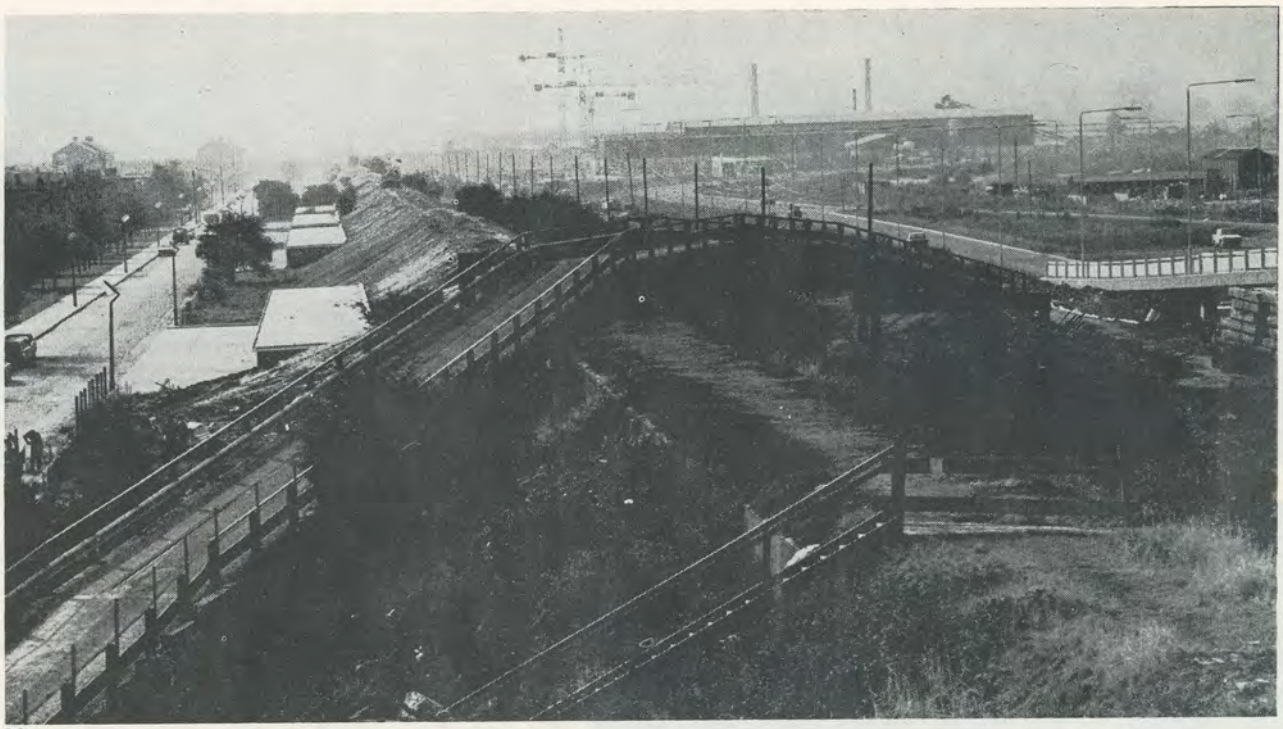
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17 *Park Hill, Sheffield. Its linear form and 'street decks' strongly influenced Thamesmead's designers*
18 *Part of model from north-west showing strong linear connection intended between first stages and centre*
19 *Town structure diagram shows high density 'skeleton' and open space system used as acoustic buffer to lower density areas*

20 *Methods of traffic noise reduction: top (for spine road), earth banks and 300ft strip of playing fields; middle (for distributor), continuous single-aspect linear housing with southerly orientation; bottom, industrial barrier where road is located to south (from 1966 report)*



21



22

distributor roads do not generate much noise and dual aspect buildings are probably satisfactory. In stage III the yardstick has determined that part of the spine building is dual aspect **27**. The consequence is the opposite of the intention; instead of protecting dwellings from noise, the spine building exposes the maximum proportion of the population in a dense strip.

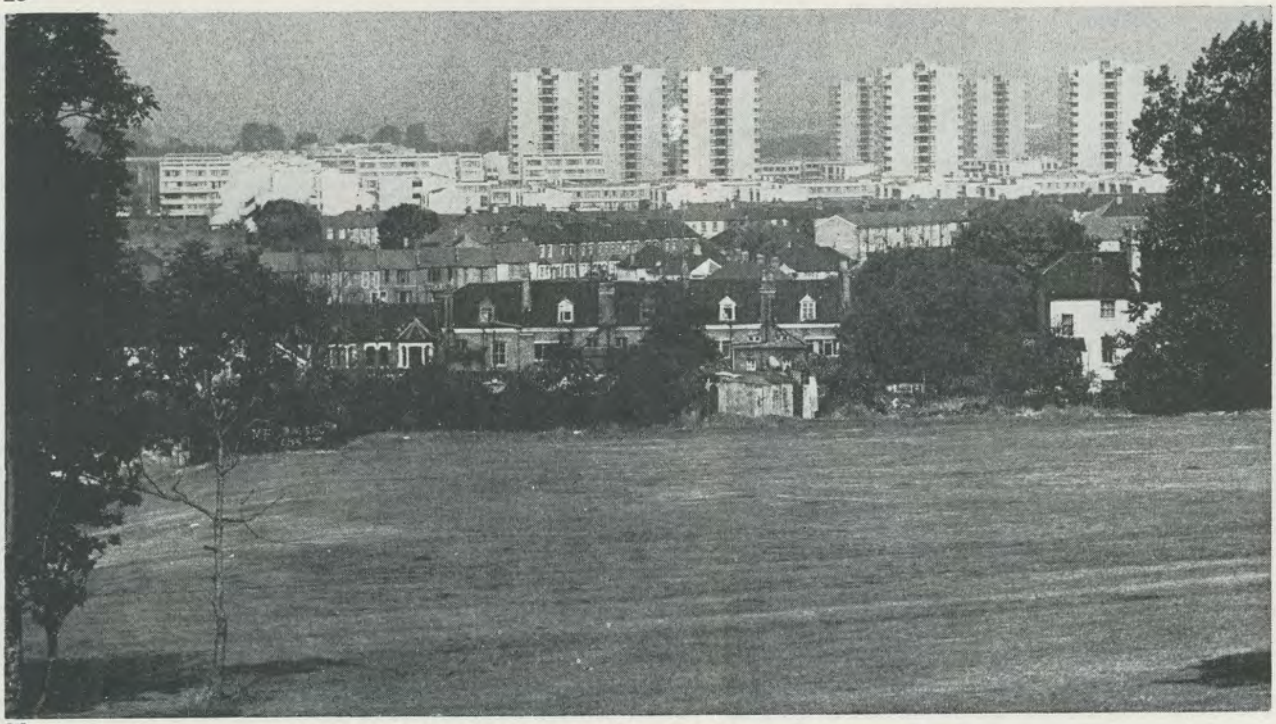
The deck system

A number of factors have contributed to the idea of the deck system contained in the spine buildings. In the first place the old by-laws of Bexley prescribed that no habitable room should be below the level of 8ft 6in above Newlyn datum, which affected the design of stages I and II. This constraint no longer applies now that a decision on the upstream Thames barrier has been made and river wall defences are to be built. The idea was sustained as an obvious device for vehicular/pedestrian segregation and as a way of getting over the sewer barrier **21** to the north. By putting cars underneath, dwellings in the spine building open off a narrow paved street and face out onto private balconies. The deck is made into an almost continuous upper ground plane which in stage I runs along the terrace houses as well. Finally, what sustained the deck in a physical sense was probably the belief that, at this density, deck access was necessary anyway and that it should be a continuous integrated system instead of a series of isolated galleries serving individual blocks.

As important as these physical factors were the social qualities which the architects believed could be effected by building form. The Woolwich-Erith report speaks of 'linking the community by high-density spines of dwellings'



23



24

which 'will give the town a basic skeleton around which the people will live' and concludes that this approach will 'bring into being a new environment and life . . . of considerable urban merit'.

Ideas of this kind were most coherently expressed by the Smithsons proposals for deck housing **15** and the cluster city **14** in the early 1950s which the *Team* 10 *primer* exposed to a wide audience in 1962. Their argument rejected the arbitrary social subdivisions of slab and tower blocks and sought forms of physical organisation, equivalent to street and district, which would really represent the new scales of association in modern society.

For sociologists the defect of this approach to urbanism is the same as that of the neighbourhood theories it set out to replace. It equates community with physical form whereas people increasingly inhabit a non-place community, identified by patterns of consumption, careers and loose social networks related to work colleagues and recreation rather than district.

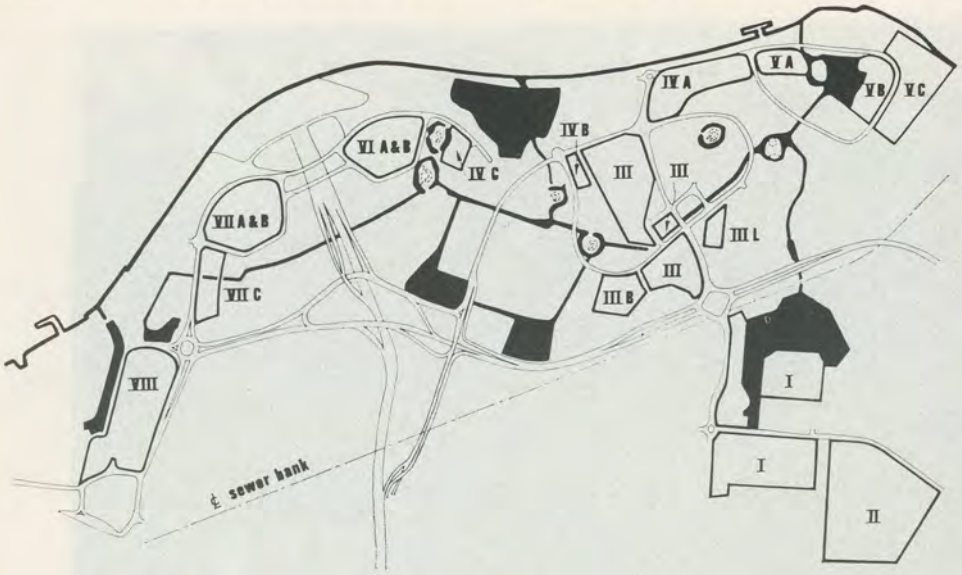
There is evidence, among community workers, that the deck system succeeds as a medium for renewing and perhaps making acquaintances, principally for women and children,

21 *Sewer bank—the great divide which will always separate first stages and Abbey Wood estate (left) from rest of town. In background,* GLC*'s Balency factory*

22 *Abbey Wood estate. Thamesmead is part of suburbia and suffers its poor communications*

23 *Prospect from suburbia—across Harrow Manor Way to stage* 1 *linear block*

24 *Image of a city in a suburban setting: view northward from slopes of Lesnes Abbey*



Programme for stages
I complete
II for completion autumn 1973
III for completion mid-1975
VIII start mid-1974
IIIb and I start 1976
IVa start 1976
Va start 1977
Vb start 1978
VIIc start 1978
IV b and c start 1979
VII a and b start 1979
Vc start 1980
VI a and b start 1980

25



26

for men have less reason for using it. But it is important to realise that this is a function of the route being limited, rather than raised, and could equally well occur on the ground.

Intensity of use depends on the facilities to which the route offers access and on its continuity. The architects have been thwarted; in stage I the brewer wanted the pub on the road not the deck; Bexley insisted upon a separate primary school associated with its playing fields; the old people's clubroom is separate; and the youth club has joined the school. Unfortunately, the vision of a village street in the air, alive with corner shops, occasional small industries, nursery schools and so on, has not been realised.

More serious is the difficulty of achieving continuity. Even within stage I the deck system is broken by Yarnton Way **30** and one detects an inherent problem in combining the spine building with the road layout. In the master plan the spine building is usually shown as a continuous band progressing towards the centre, but in its detailed working out the highway requirements can explode and offset the connections. Stages I and III are separated by the junction between the Woolwich/Erith road and Harrow Manor Way **28**, and the connection between the two parts of stage III coincides with a local distributor junction.

Within stage III the yardstick has had its emasculating effect and the main pedestrian deck fails to cross the access roads which penetrate the spine **27**. This seems a disaster; either the deck system should be sustained as a convenient route or the principle should be abandoned.

The pedestrian system through open space

An alternative to the deck is a system of paths which will pass through the low-density housing areas and public open spaces and along the canals. For some of the residents in the lower-density areas, this will be a more direct route to the centre now that it is clustered on a north/south axis rather than arranged along the river. The architects have described what they hope will happen on these routes: 'stopping to rest; looking at surroundings, river, sunset; watching children at play'. It promises to be delightful and the pleasures are already appreciable at stage I where the lake and exceptional landscaping make the kind of place where it is good to be on a Sunday afternoon in summer **31**.

Density and building form

The planned density for Thamesmead's housing was originally 100 ppa with a proportion of 65 per cent public housing to 35 per cent private giving a total of 60 000 people on 600 acres of housing land.

The Thamesmead reappraisal, published by the GLC earlier this year, revealed that private developers would be unlikely to build at a density higher than 15 houses per acre and that the occupancy rate for private houses was much lower than the one person per habitable room assumed for public housing. Thirty-five per cent of private housing will now take up 50 per cent of the land and bring down the total population to below 50 000. With public housing remaining at 100 ppa there will be rather obvious distinctions between the private and public sector—an urban environment for the workers with the majority in flats and maisonettes and suburbia for the owner occupiers.

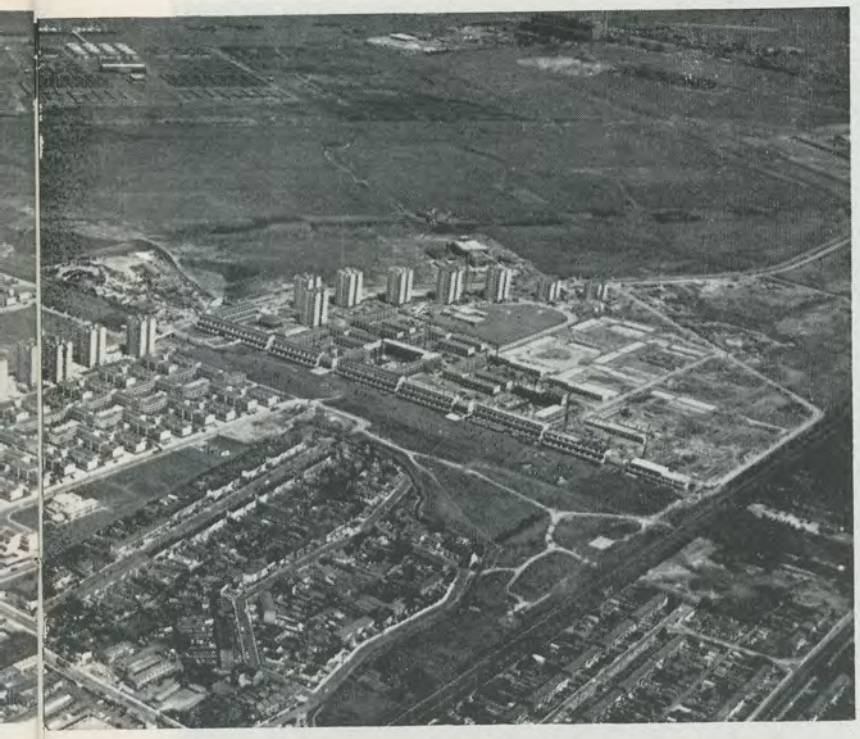
It is worth briefly considering certain ideas about building form which could help blur these distinctions. The density of 100 ppa is an interesting one, being about the maximum which allows terraces with the suburban amenities of private gardens, integral garaging and a proportion of public open space. The public housing at Thamesmead could be the same kind as that offered in the private market even though the frontages would be less.

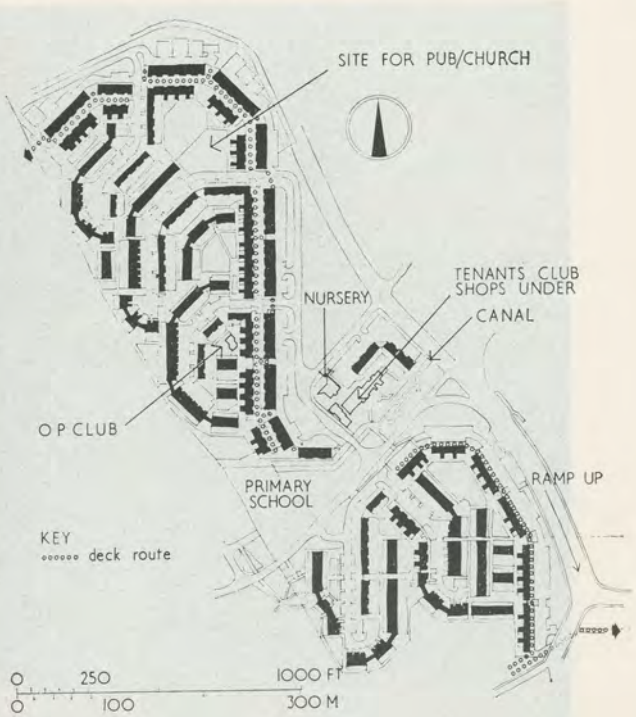
Another Thamesmead statistic which is relevant to the pattern of land use, though less conclusive, is the ratio of residential land to open space. The combined acreage of recreational and educational land to open space in the master plan, before reappraisal, was about 500, not far short of the area of housing itself. What is interesting about this, as Sir Leslie Martin has demonstrated*, is that as the proportion of open space increases in relation to housing, the housing form can be less constrained without loss of density. For example housing and school land are usually arranged separately, but if the housing is converted into a ribbon around the school **32** and the school playing fields are dual use, views are opened up and private gardens have





*Education without walls. RIBA *Journal*, 1968, August, p356

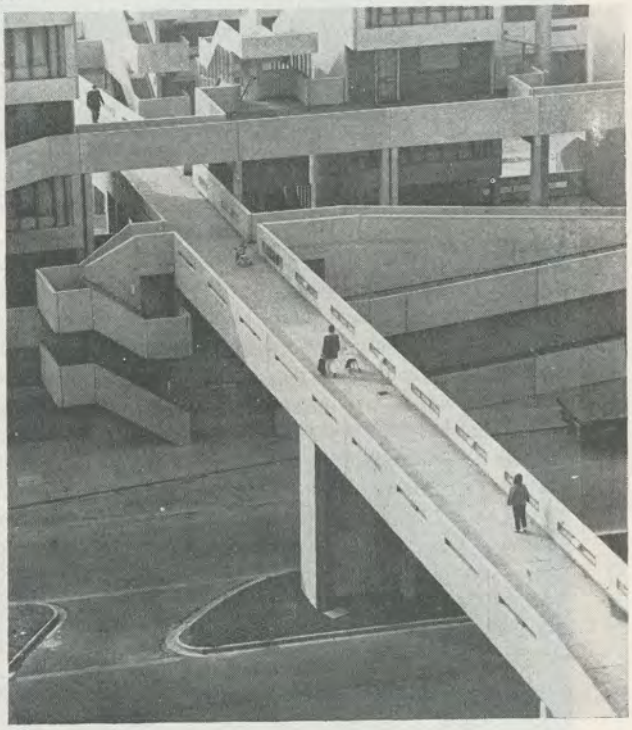




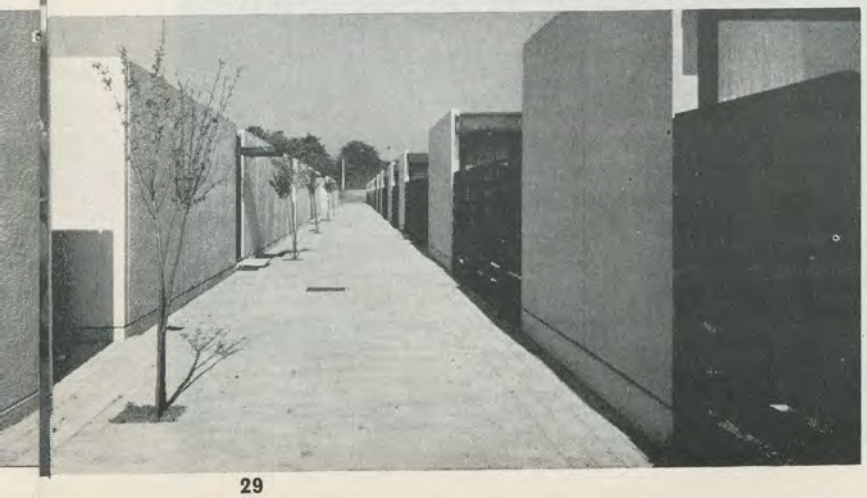
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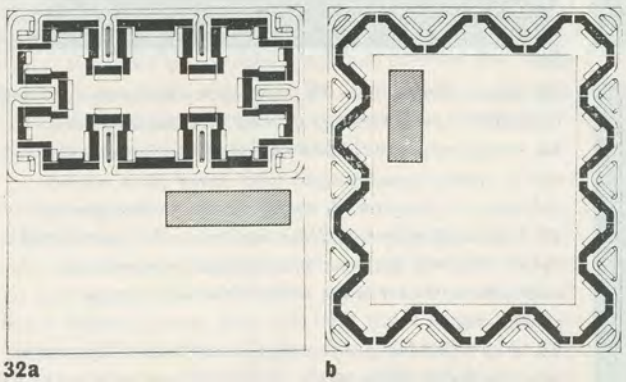
25 *Latest staging plan for local authority housing. Compare with* **10** *for modifications to road and canal systems.*
26 *Aerial view* (1971) *from south-west, showing stages* I *and* II *(under construction) with linear park between: suburbia in foreground, sewage works in background*
27 *Layout of stage* III. *Note that while the 'spine' still exists, its form has been emasculated by yardstick restrictions, and that the raised deck is no longer continuous*
28 *'The spine buildings progress northward like trains towards the future centre'*
29 *Albertslund housing scheme, Denmark (architects V. Moeller-Jensen, T. Arnfred, M. J. Pedersen and J. O. Soerensen)*
30 *Yarnton way divides stage* I *in two parts. Note how much visible structure is devoted to pedestrian movement*



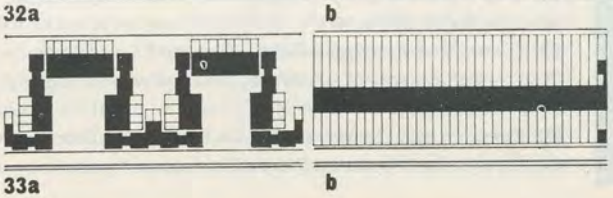
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access into a park. A further development of this argument, which needs investigation, is that because of the large area of recreational land adjacent, it may no longer be necessary to provide public open space within the residential area. With little loss of density private plot sizes can be increased by absorbing a greater proportion of the residential band into private ownership **33**. The argument advanced by the GLC against using recreational land to disperse housing is that much of the land is already allocated as an acoustic buffer along the spine road. This land would be freed if the insulation were achieved instead by an earth embankment. Studies of the kind illustrated suggest an environment with a less dramatic image than Thamesmead which could be entirely in three-storey housing at 12ft frontage or 75 per cent at 16ft frontage. The GLC have argued that such an approach would be 'oppressive' in the flat landscape of Thamesmead. This assumes that the physical image has greater significance than the amenity which terrace houses and gardens gain over flats and maisonettes, and the levelling of social distinctions which a multi-class low-rise scheme implies. Anyway, Amsterdam is a good place and the ingredients are basically the same.

Thamesmead belongs to the period of Willmott and Young's *Family and kinship in East London*, Jane Jacob's *The death and life of great American cities* and Roger Mayne's photography of the East End. It represents a reaction by architects against the middle-class values of suburbia and the feeble aesthetics of the mark I New Towns in favour of a bold act of communal architecture which conceives the town as a building and preserves the values of street life. The intention is noble and the result in stage I, spared the discipline of the yardstick, is spectacular and highly accomplished **35** if the premises are accepted. But, with the easy wisdom of hindsight, it is difficult not to feel that something altogether more modest and private might have been achieved, appropriate to a potentially suburban situation, without betraying social and aesthetic ideals. The master plan represents an image of a high-density town rather than a strategy, and this commitment is becoming increasingly difficult to sustain as it is assaulted by the yardstick, changing attitudes about community and new facts about housing form. The plan also assumes the public authority distanced from the user in order to speculate about his needs and engineer his social environment. Over the next decade this situation must change as the paternalism of local authorities is challenged by tenants associations and co-operatives articulate about their own values.



32a b



33a b

31 *A pleasant place to be on a summer afternoon*
32a *Diagram of three-storey housing at* 100 *ppa laid out adjacent to school (both sites* 17 *acres);* **b** *Still the same densities and acreages, but housing arranged as ribbon around school playing fields (R. MacCormac)*
33 *With a larger area of recreational land adjacent, individual plot sizes as* **a** *may be increased to take in all communal housing land* **b** *with little loss of density*
34, 35 *Not 'before' and 'after', but a contrast of images. Should perhaps Thamesmead's fall somewhere between the two?*

PHOTOGRAPHS

All photographs by Tim Street-Porter except:
11, 13, 18, 26, 31 by GLC photographic unit, Department of Architecture and Civic Design
17 by Arthur Winter
29 by courtesy of the Royal Danish Ministry for Foreign Affairs



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Building illustrated

CI/SfB 054

New town
GLC Department of Architecture and Civic Design

Thamesmead: part two



RICHARD MACCORMAC *concludes his study of the* GLC's *1100 acre riverside project with an examination of the first two stages, now almost complete, designed by the council's architects*

ARCHITECT'S ACCOUNT STAGES I AND II

The detailed design for stages I and II was, to a large extent, developed at the same time as the master plan and implements the master plan principles of pedestrian/vehicular segregation **A**. Upper level pedestrian routes, with their system of pedestrian bridges, follow the major traffic routes. Low-rise housing areas are based on a system of ground level pedestrian circulation, uninterrupted by vehicular traffic. The linear block running alongside Harrow Manor Way **4** (the future principal traffic route), though no different in terms of population density from other parts of the site, nevertheless forms a visual beginning of the system of high density, high-rise spines envisaged in the master plan. The close integration of housing with green spaces and water **6**, **7** will certainly be echoed in the later stages of the development.

The gross site areas of stages I and II are 85 and 127 acres, with respectively 50 and 41 acres available for housing. This, at a density of approximately 100 ppa, produced 1490 dwellings in stage I and 1251 dwellings in stage II.

The client required the maximum number of family dwellings to be designed as individual houses with gardens, and in both stages well over a third of the dwellings are provided in that form. These are primarily four- and five-person houses **O** with a smaller number designed for six persons **P**. There are also some four and five-person maisonettes at first and third-floor levels of the linear block **C-K**, **L-N** and in those the absence of a garden is to some extent compensated by a very generous balcony-terrace adjoining the living room.

There are some three-person flats, designed for old people, accessible from the first-floor level pedestrian deck. These are in addition to two blocks of one- and two-person flats for old people linked to the linear block and to its pedestrian deck **28**. The aim here was to integrate (as much as possible) the older inhabitants with the rest of the community without exposing them to the noise and other forms of nuisance so often generated by children from larger family dwelling units.

Finally, small dwellings for two or three people are located in the 13-storey point blocks **B**, 12 of which have been constructed in stage I and a further 9 in stage II. The client required these small dwellings to be sited away from larger family units. At this density it was considered that they would be best situated in high-rise point blocks.

The distribution of various sizes of dwellings in different types of blocks is (with only minor modifications) the same in both stages, although in stage II both the layout and the buildings have been somewhat simplified.

The local by-law in force when the scheme was being designed required that there should be no habitable rooms at ground level. As a result the whole scheme was raised off the ground, the ground floor being used only for garages, tenants' stores, services and, in the case of three-storey houses, for access to dwellings. This particular by-law requirement, while obviously increasing the costs, provided an opportunity for hiding the garages, stores etc under the buildings and made the principle of upper level pedestrian circulation easier to implement.

Although stage I is located immediately next to existing housing areas with existing shopping and community facilities, it was considered essential to the early success of the scheme to ensure that these facilities are available within the development as soon as possible after the first tenants have arrived. A local centre in stage I now contains 10 shops, a bank, a police office, and a rent office as well as a specially designed medical centre and a tenants' club-room. A smaller local centre in stage II contains tenants'

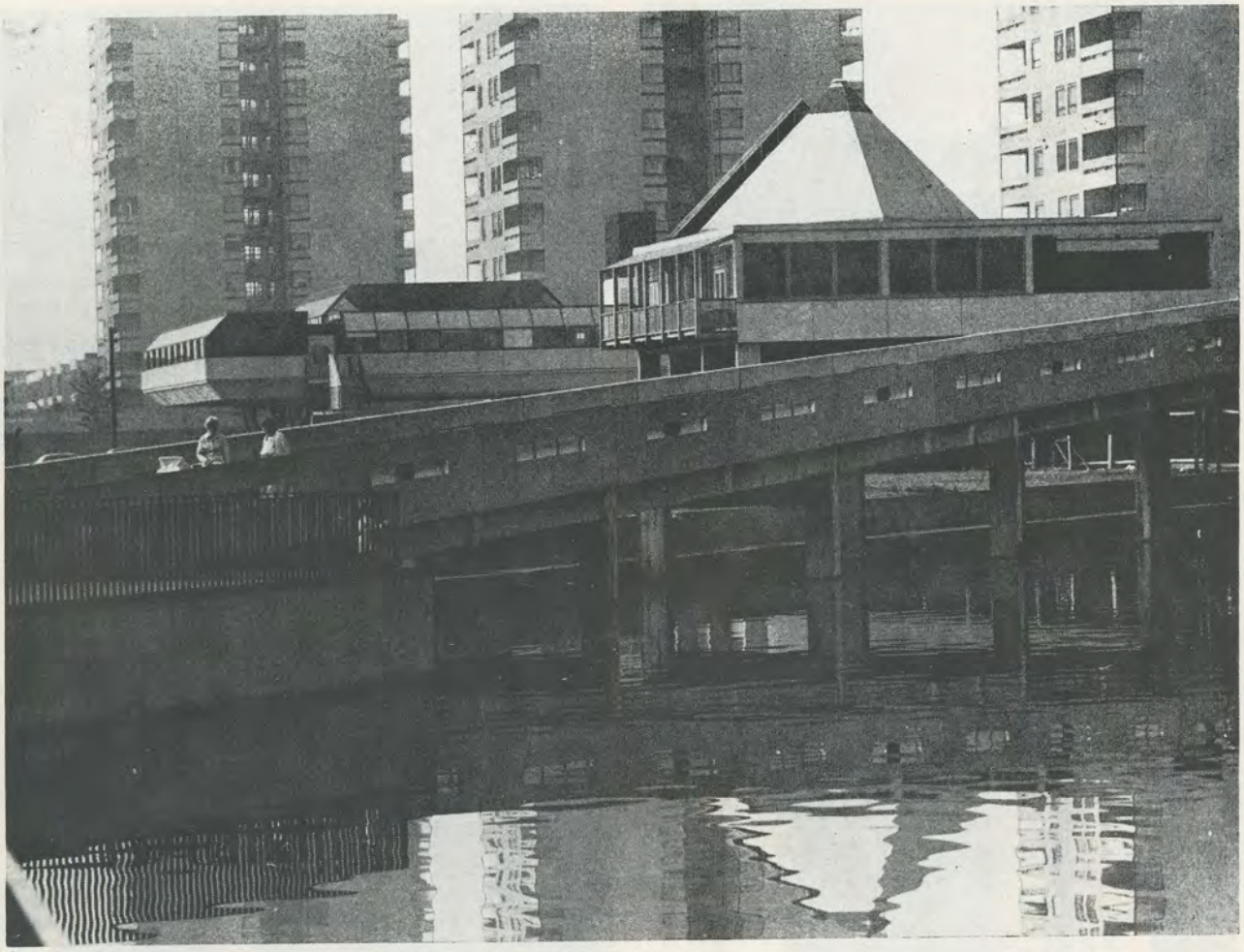


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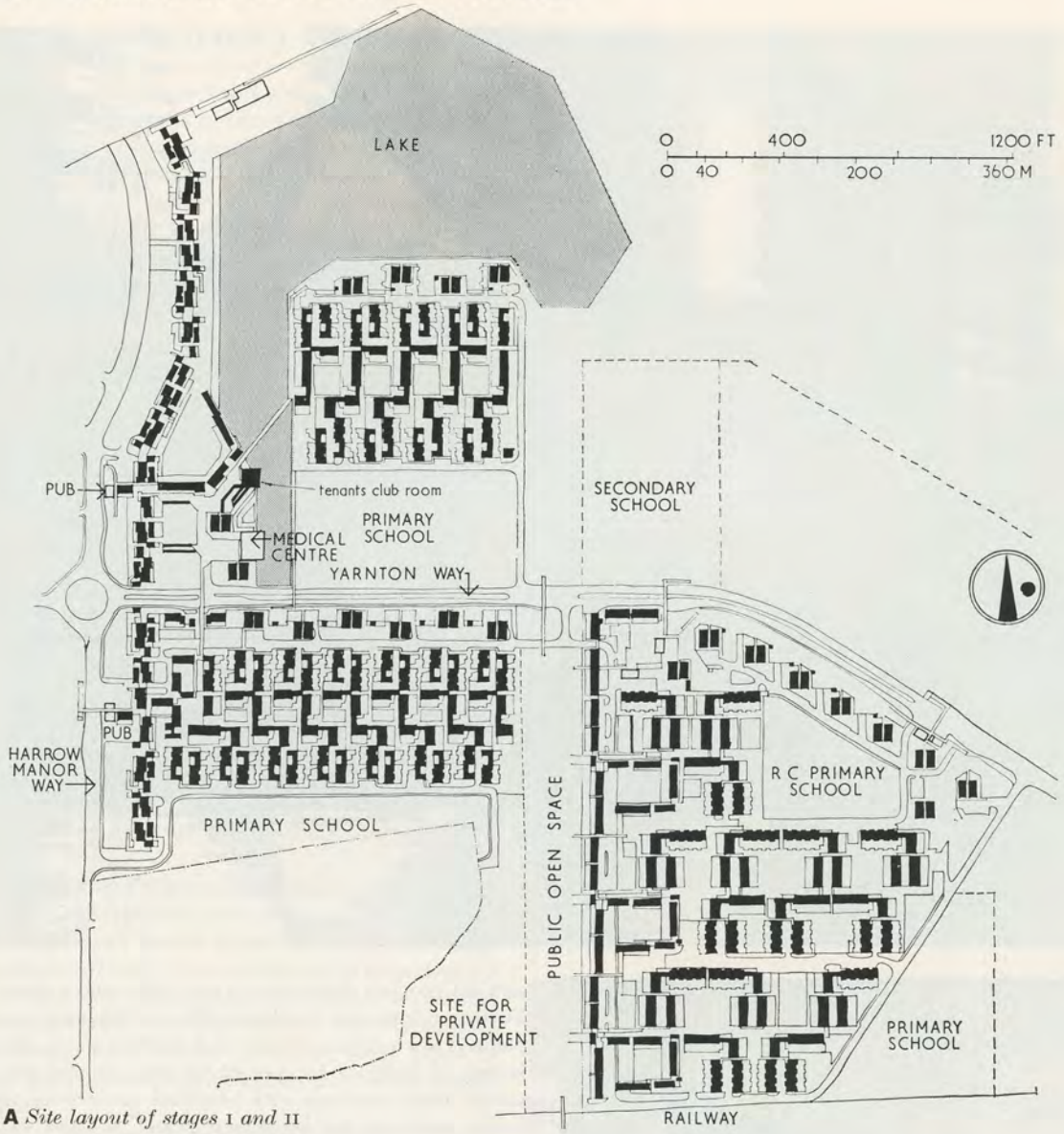
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and old people's clubrooms, a rent office and a shop.

Further community facilities either under construction or planned are a branch library and meeting room, two public houses, a building for pre-school play groups and junior youth, three churches with attached community rooms, a youth centre in the secondary school, a water recreation area for children, public lavatories, boat storage for school sailing classes, a lakeside restaurant, boatstore etc. The surrounding public open space also provides sports pitches, adventure playground and one-o'clock clubs.

In addition to the amenity open spaces immediately next to the housing and containing a number of small play areas for toddlers and juniors, a public open space link **18** has been created between stages I and II joining the woodland of Lesnes Abbey with the first stages of the development, the lake and the whole future system of public open spaces throughout Thamesmead. The 25-acre lake also forms part of the public open space area. Its recreational potential is obvious and very extensively used **21**, but it is not always realised that it is also an important element in the Thamesmead land drainage system, storing the surface waters

1 *Abbey Wood estate: 'people's vernacular' of early 1950s*
2 *Architecture of stage I 'is persistent—sharp and invigorating or tiresome and obtrusive . . . depending on your state of mind'*
3 *Bridge over lake with tenants' clubroom and health centre beyond—'an artificial environment of streets and alleyways enclosed by concrete'*
4 *Stage I is separated by linear block and north-south distributor road from encroachment of suburban fantasy*
5 *To south-east, inter-war semis—drab but adequate*



A *Site layout of stages* I *and* II

collected from the surrounding areas until, at low tide, they can be discharged through the system of canals into the Thames (see plan AJ 11.10.72 p828).

The provision of schools in stages I and II is the responsibility of the London Borough of Bexley and it is perhaps symptomatic of the sort of close day-to-day co-operation which exists between the GLC and that local authority that the primary and secondary schools have been completed at the same time or even before the housing areas which they are to serve **8**. There are now four primary schools in stages I and II (one of them Roman Catholic) as well as a large secondary school situated on 36 acres of land.

The construction of stages I and II is being largely carried out by industrialised building methods. The system adopted is that developed by the French firm of Balency at Schuhl and a special factory has been built on site to produce the precast concrete elements. Only the linear block and old peoples blocks in stage I and the special buildings (such as shops, clubrooms etc) in both stages are built using the traditional methods. White cement and white aggregate are used on the faces of external concrete units to minimise the feeling of drabness so often associated with large panel concrete structures. This, with the extensive use of facing bricks in the public areas, natural wood in the form of cedar window mullions, and the articulation and modelling of the facades, gives the scheme a human scale and a warmer, more intimate character.

APPRAISAL

by Richard MacCormac

Driving from London, the first buildings of Thamesmead to come into view from the new Woolwich—Erith road are the spine blocks **4** of stage I, which immediately establish the town's distinct identity in contrast to the surrounding area. To the south the housing is a typical outer London mixture of inter-war semis and Victorian terraces, drab but adequate **5**. To the west the GLC's Abbey Wood Estate, in the LCC's pitched roof and brickwork people's vernacular of the early 50s, with grass and tidy front gardens presents a polite and vacuous kind of suburbia **1**.

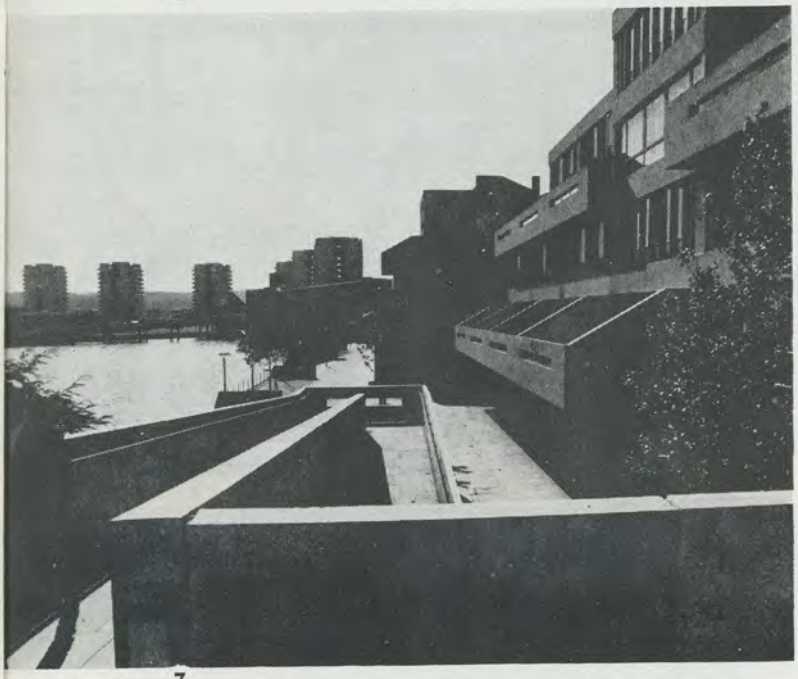
The new Thamesmead buildings stand out as a spectacle which demands your attention and response before you turn away. The architecture is persistent—sharp and invigorating or tiresome and obtrusive, depending on the weather, or your state of mind **2**. It announces uncompromisingly that this is a new community whose surroundings will be an artificial environment of streets and alleyways enclosed by concrete **3**, and symbolically separated by the spine block and the north-south distributor road from the encroachment of suburban fantasy.

Planning structure

Stages I and II **A** are not typical of the general form of Thamesmead because of their separate position south of



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the sewer bank. The environmental areas along the river will be defined by the district distributor along their north edge where the higher densities occur and decline into lower densities towards the south to fringe the linear park. Stage I is defined on the north side by the east-west spine road and along the west side by the existing position of Harrow Manor Way and split in two by the distributor road (Yarnton Way) which bounds stage II. The position of this road, coinciding with the existing alignment of Eynsham Drive, divides stage I into two distinct parts and severs the northern higher rented housing, already isolated on three sides by water and open space, from the southern area of public housing.

The distribution of open space in stage I is distinguished by the balancing lake, located in the lowest part of the site, which drains the surrounding area and discharges water via the canals at low tide through one-way valves at Cross Ness. The lake penetrates the housing and fronts the spine block **6**, **7** to give it the same watery prospect as the future spine building along the river front.

In other respects the location of open space in stage I conforms to the general concept of the master plan. The primary schools **8** are inserted in fingers of space opening off a greenway at the east edge to which the low-rise housing has access **17**.

The intentions behind the structure of stage II are less distinct and its context is even less characteristic than that of stage I. It lacks the advantage of the lively effect of water, but to compensate for this the spine building has a view over the greenway. This must be delightful for its residents but contradicts the principle of density structure adhered to elsewhere. What makes this unsatisfactory in terms of the master plan argument is that the low-rise housing and one of the primary schools associated with it abutts land safeguarded for industry, when one might have expected a spine building to perform the task of visual and acoustic protection. Another inconsistency is that the spine building does not continue towards the town centre, but stops short. It is interesting to note that the master plan did not propose any high density housing in stage II, suggesting that it might be considered an appendage of stage I. (See AJ 11.10.72 for analysis of master plan.)

6 *Northern section of stage* I *linear block overlooks lake: view north towards temporary boiler house and* **7** *south towards local centre*

8 *Primary school by London Borough of Bexley*

9 *Tower blocks in stage* I *cut off by access road from low-rise housing*

The road system

The distributors related to stage I and II will remain to some extent isolated from the main distributor system of the towns and the loadings will probably not exceed 00000 vehicles an hour. Nevertheless, these are fast roads and their isolation from the environmental areas *feels* appropriate even if one disagrees, as I have done in my previous article, with the measures which achieved this.

Traffic swishes past on Yarnton Way which is not pleasant to cross, and one feels exposed on the limited width of pavement. In contrast, within the southern area of stage I, pedestrians generally take precedence and the pace and sound of traffic is successfully diminished.

Access roads

In stage I the access roads are placed at the edges of the low-rise housing areas and run along the west side of the northern part of the spine block. In these positions they do not sever the pedestrian movement within the housing areas and towards the shops and local centre but they do, to a small extent, intercept the north-south movement between housing and the two primary schools which could well have been within the mini environmental areas defined by the access roads. The two rows of tower blocks are also severed from the low-rise housing **9**—not inappropriately as their occupants, mainly without children, are in each case potentially cut off from the noisy children's environment to the south.

In stage II the access roads into the low-rise area are rather differently arranged as culs-de-sac running east-west, and subdividing the housing into strips. These roads are long and straight, encouraging vehicle speeds incompatible with the 'soft' green spaces within the housing courts intended for toddlers, which are exposed to their noise and danger.

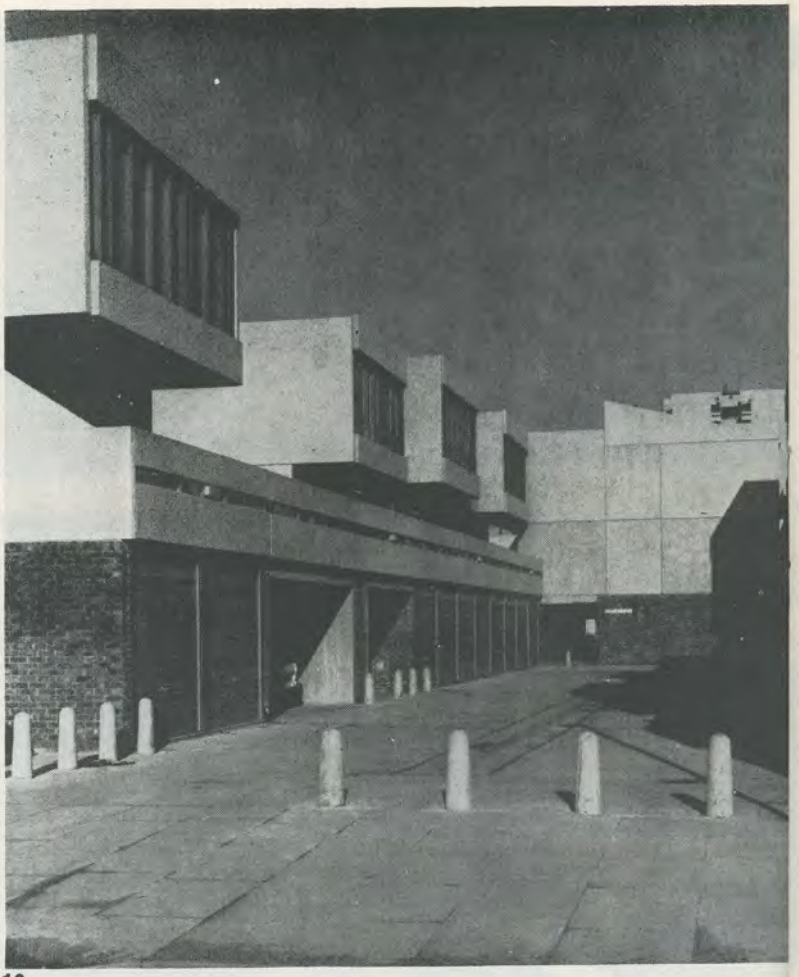
Culs-de-sac and car parking

One of the difficulties encountered in designing higher density housing is the greater proportion of space taken up by car parking and the road hierarchy. In the case of the tower blocks and the spine buildings this is solved by decking over the garages and the roads serving them. This is effective in conserving space, but inherently bleak and disproportionately expensive and wasteful of resources given the yardstick for this density. The proof, in the low-rise parts of stages I and II, is that terrace housing can meet this problem without resort to such gestures.

The culs-de-sac in stage I are exceptionally narrow **10**, 26ft from face to face of the houses, a dimension which is not usually permitted by planners on grounds of privacy but which seems to be acceptable to the majority of tenants. Separate pavements have been discarded and the scale is tight, like a mews in which completely hard surfaces are acceptable, and the atmosphere is far better than the decked-over parking. The provision for first-floor access allows two car spaces per dwelling, but as only a proportion of the houses back onto culs-de-sac, the total percentage of car provision within them is about 100 per cent; additional space is required beside the access roads for visitors.

Pedestrian routes, recreational space and children's play, and the spine block deck system

In the first part of this study (AJ 11.10.72 p826) I looked at the factors which generated the deck and concluded that it was a dubious solution to the problems stated. But it is well worth looking at because, although it may be unreasonable, it largely succeeds in realising its subjective intentions. The unreasonableness is an important ingredient, for logically the deck should be straight like the access roadway



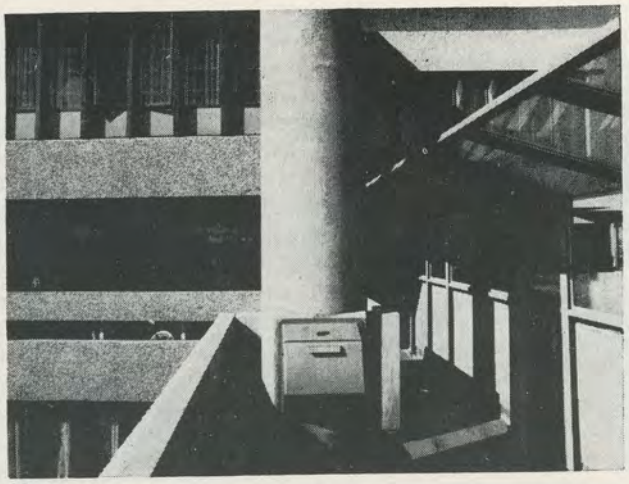
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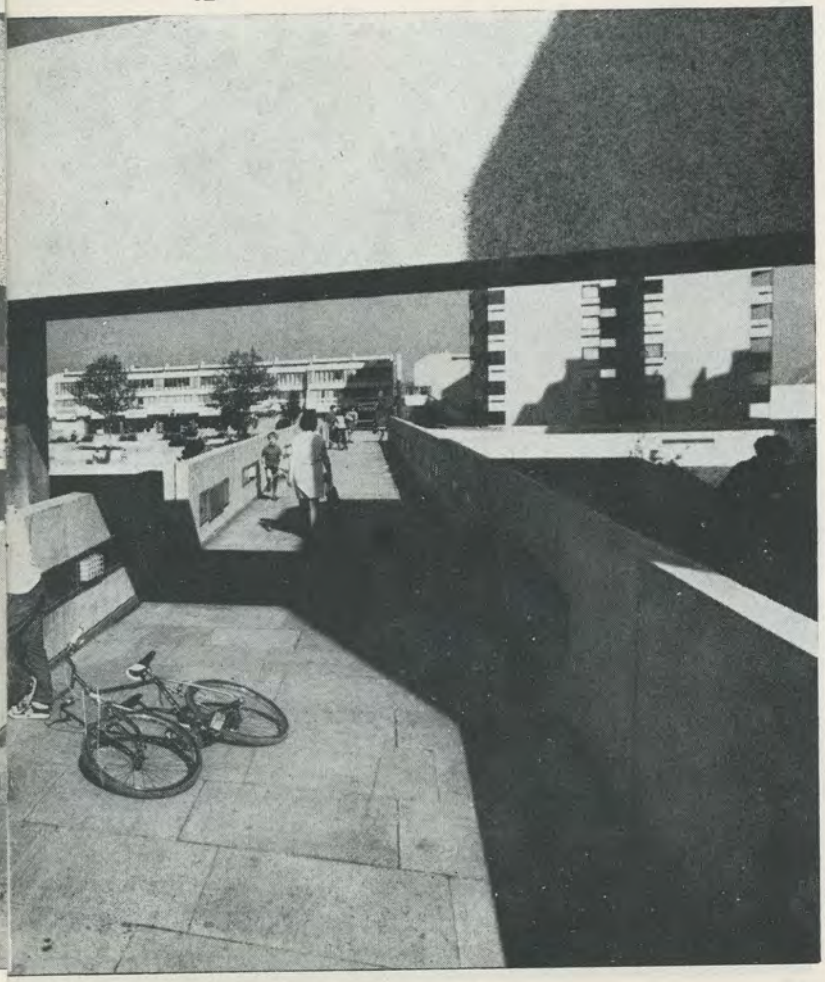
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underneath and the only purpose of the alternating permutations of dwelling types is to effect a constantly changing pedestrian environment. This is a place which excites people and which the rationale of modern architecture does not produce **12.** It epitomises a dilemma for architects for on the one hand we sense that people enjoy this kind of thing and on the other we are slightly disgusted that they respond to the derivative and picturesque rather than to the products of logic. The deck is a medieval street and the architects are not ashamed to say that as well as Parkhill and Habitat, Thomas Sharpe's *Anatomy of the village* was a source of imagery during the design.

The village idea makes itself felt in a number of ways. It is good to find front doors left open and to have the street affected by music and chatter from inside the dwellings—reassuring domestic sounds which we find when we go on holiday to old villages by the sea. But it's a pity, in terms of this particular fantasy, that living rooms or kitchens do not front onto the street.

With the front doors open, children scamper along the deck from house to house, leave their toys outside, ride bicycles and chase dogs, and people stop to chat in the way that was intended. Under foot the surface alternates from paving to brick, fortuitously, so the quality of the street is subtly altered as you walk. But how strange it is that it should be

10 *Culs-de-sac in stage* 1 *are narrow, mews-like in atmosphere, with acceptable hard surfaces*
11 *Bridge across Yarnton Way leading north to* **14** *local centre*
12 *First-floor spine like a medieval street—something the rationale of modern architecture cannot produce*
13 *Windscreens on upper galleries 'are acquiring the patina of old bus shelters'*

necessary to fall back upon devices of this kind, which reflect the disorganisation of pre-industrial society, in order to disguise our own logic and administrative capability!

Access in the stage I spine blocks is not entirely charming in this way. The galleries connecting the upper level maisonettes and flats are rudimentary, with asphalt underfoot and elegantly detailed glazed wind screens which are acquiring the patina of old bus shelters **13**, only partially protecting the pedestrians from the vicious winter winds which blow down the Thames. However seductive the subjective quality, deck access remains inconvenient and draughty both as a means of private access from car to dwellings and as a means of access to public destinations. In stage II economy has ruthlessly simplified the section so that the deck in the spine building runs for 1600ft on the east side of the building. No longer a village street, this reformed version evokes Corb's references to the promenade deck of the *Aquitania*—and it's boring.

The pedestrian environment in the low-rise housing

The now obsolete Bexley by-law, which required all housing accommodation in stages I and II to be 8ft 6in above Newlyn Datum suggested the deck access idea, not only for the stacked dwellings, but for the low-rise houses too. The architects originally envisaged a continuous system on first-floor level which would enable pedestrians to reach all destinations within stage I without descending to the ground. Eventually this proved too expensive to execute and the terrace houses are only linked in groups across the site at right angles to the principal direction of movement towards the spine block and the centre, the bus stop in Harrow Manor Way and the bridge across Yarnton Way **11** to the local centre **14**.

So the upper level system does not offer a direct route which avoids stairs, for mothers with children, prams and shopping, which should have been its justification. Its practical advantage, already mentioned, is that the space consuming slots leading to front doors between garages, usually necessary to three-storey housing, can be taken up with cars. Its subjective advantage is to complicate the environment and make it labyrinthine and explorable, particularly for children, to whom it offers another dimension for chasing games. On the other hand it is confusing, for the houses perceived as groups at ground level are not necessarily connected at deck level and consequently the house numbering is difficult for visitors to grasp until they notice the colour coding of doors and gates. The housing manager was originally very unhappy about the deck and the architects agreed that if it proved unpopular they would convert it into private balconies. The secretary of the tenants' association says that the residents do not feel that the deck violates their privacy. This is surprising, for visitors may feel that they are continuously trespassing on personal territories and dislike the free view down into private gardens.

The deck fulfils the rather special values of Thamesmead which put community before privacy. The individual identity of the houses is subordinate and it is difficult to imagine them in private ownership.

The main pedestrian areas at ground level in the stage I low-rise housing run east-west, connecting the spine at one end and the linear park at the other. The feeling of moving in from the access roads and culs-de-sac at the edge of the housing to the vehicle-free oasis at the centre **15** is very pleasant and the accompanying change from the tight scale of pedestrian alleys and mews to the series of courts along the centre is good too. Visually, one responds to the intention of a tightly knit urban environment which is so much better resolved, in its own terms, than the litter of terrace



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15 *Feeling of moving away from vehicles into a pedestrian environment is very pleasant*
16 *'Incessant and inescapable noise of children' dominates low-rise area*
17 *East-west pedestrian ways offer glimpse of parkland through succession of openings*
18 *Grass ridge stretches south towards Lesnes Abbey. Recreational areas are pleasant enough to bring people out just for a stroll* **19**. *Stage* II *linear block under construction left*
20 *Gardens are important social link which maisonettes deny*

houses along access roads in the Abbey Wood Estate. Even so, one has reservations about this urban rather than suburban scale. Where it is too small for grass, the hard surfaces make it acoustically harsh. The incessant and inescapable noise of children dominates the spaces between buildings during the evenings, holidays and weekends **16**.

Stage II is more open and the pattern simpler, but it will probably be less easy for the pedestrian to grasp, because the access roads do not define the edges of the housing areas but penetrate it like the pedestrian ways and the housing form is the same in each case. There are even pedestrian culs-de-sac off the east-west pedestrian ways—paved and used by children—which are little different from those serving cars.

It would be interesting to compare the social milieu of the low rise housing with that of the spine block in stage I. The latter consciously attempts to achieve the neighbourly qualities of street life by removing the road from between the houses. On the other hand the building form which achieves this denies other media for social interaction which exist in the low-rise situation **20**. There is no house-to-house link on the garden side to give children access to each other without going through the house; kitchens do not overlook the street so the passing faces do not become familiar; and the garages are completely detached from the dwellings, so that car cleaning rituals which can be a social focus in a cul-de-sac are likely to take place amongst strangers.

Recreational space

One of the best things about the east-west pedestrian ways in stage I is the glimpse they offer, through a succession of openings, of the open space on the east side **17**. This is a raised grass mound (made up of excavated material from the lake) which stretches south **18** towards the park and Lesnes Abbey, with which it will eventually be connected by a bridge over Abbey Road. This is a marvellous concept. The connection with the park is highly effective and the raised open space dramatic and unexpected. Walking along the top gives a feeling of exhilaration and children race up and down the grass slopes.

The other amenity associated with stage I is, of course, the lake **21** on which it is possible to sail or row. Promenades along the edge of the lake link with the shopping centre and the linear park, so it is possible to walk around the stage I housing area experiencing an unusual variety of environments with virtually no conflict with vehicles. One gets the impression at Thamesmead that these open spaces are interesting enough to bring people out to stroll or relax **19**, like Primrose Hill or the river front at Richmond.

The local centre in stage I

All the dwellings in the immediate vicinity of the local centre, the spine block, tower and old people's flats are at first-floor level or above, and the bridge over Yarnton Way, from the southern part of stage I, also introduces an upper level. So the shops are also located at first-floor level and service access is introduced underneath, with garaging for the adjacent tower blocks under a forecourt which is perforated by light wells ingeniously cut out of the interstices between the pedestrian routes **23**, **24** criss-crossing the space. From ground level, by Yarnton Way, the court is reached by a seemingly ceremonial flight of steps **14** across the full width of the forecourt which probably helps to give a feeling of accessibility to the upper level—although it seems a massive gesture for the benefit of people coming from the bus stop only.

The good qualities of the centre are due to its surroundings and the approaches to it—over the lake, by the future

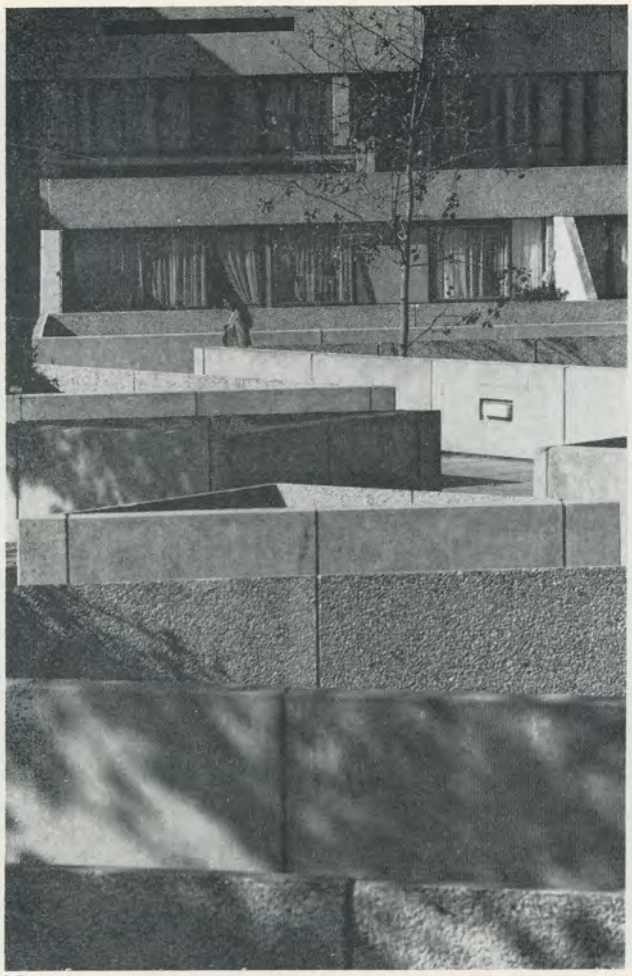


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children's paddling pool, past Derek Stow's distinguished health centre **3**, and through the unexpected and pleasant grassed court **28** behind the shops onto which the old people's dwellings look.

The dubious aspect is the complexity of structure for such a small place and the subterranean environment of the parking, a refuge for derelict cars, shop litter and children kicking Coca-Cola tins **29**. The scale and the techniques are those of a multilevel urban centre, of which it attempts to be a microcosm.

Dwelling types: the point blocks

As the architect points out (p880), the high-rise buildings were a response to the housing manager's requirement that the majority of small dwellings, two- and three-person, should be sited away from the larger noisier family dwellings. On the site plans of stages I and II the 13-storey point blocks stand in lines, as though they have been directly substituted for spine buildings.

One feels that they abruptly contradict the general intentions of the master plan by their visual independence representing a social independence—or isolation. Yet this may not be inappropriate; the small and presumably childless families in them are likely to be more mobile and less socially dependent upon propinquity than those with children. The point block offers a quiet environment, with

21 *Sailing on lake*
22 *Southern approach to local centre*
23 *Forecourt to shops looking towards spine with* **24**, *perforated light walls to service area below*
25 *Sign rather crudely related to balcony fronts over shops*
26 *Steps up to centre from old people's court behind shops*

views, perhaps of the Thames, and more civilised access than an open gallery, if the lift is respected and decently maintained.

The point blocks **B** (which are based on the standard GLC design 'PFG') make obvious use of the Balency system, with floor-by-floor repetition of a limited number of wall components which elegantly and sharply express the quality of their assembly **30**. They are popular with the site labour force because the floor by floor progress of construction has a rhythm and rate which can gradually be improved as operations are repeated and skills acquired.

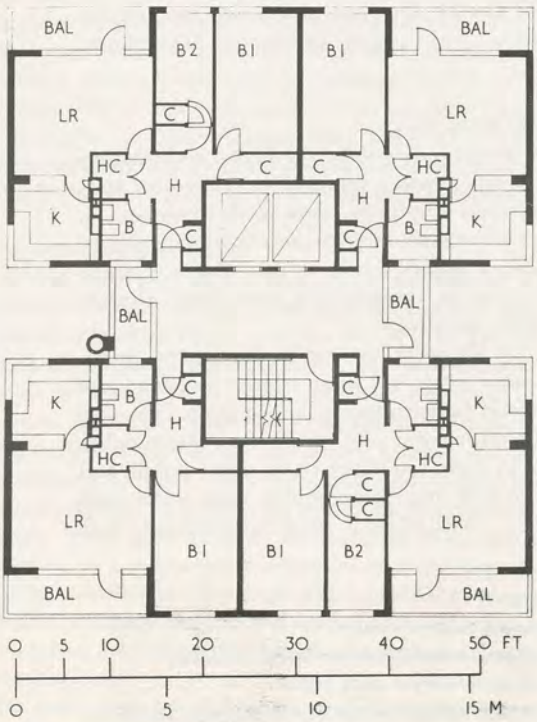
External wall panels are load bearing, with windows made as openings within the panel or at the corners of the building. Floors are cast in situ to give effective jointing between the floors and tops of the vertical units and structural continuity over them. This combination of precast walls and in situ floors, which distinguishes Balency from most other heavy concrete systems, has needed only minor modification in the design of the reinforcement in the floor slabs to conform with the standards imposed after the Ronan Point disaster.

The exterior design displays some ingenious and quirky details, such as the continuous re-entrant angles of steel which run vertically up the corners of the building holding the glazed wind screens to the balustrades, and the horizontal slots between the balcony balustrade components, which become, in the middle of the entrance elevations, ventilators to kitchen cupboards **31**. At the top the precast facing panels project up, establishing that the building is composed of planes, and the corner kitchens become little glazed boxes.

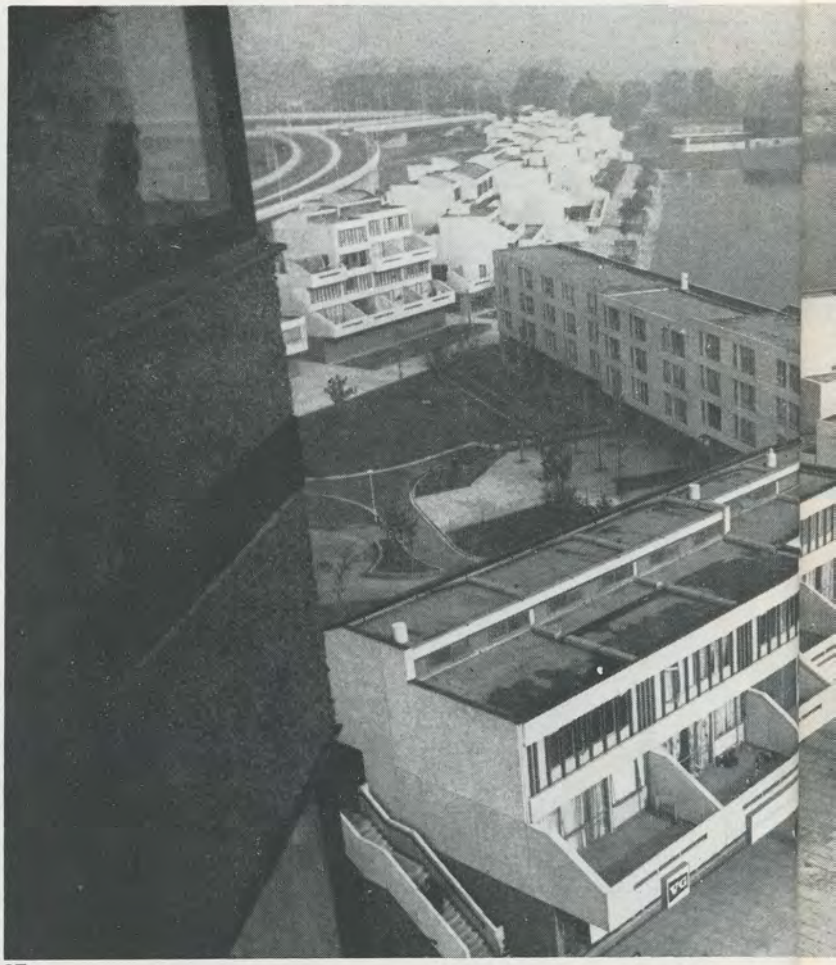
All this is enjoyable but the environment on the ground suffers. Wind speeds in the shopping centre and at the edges of the housing are increased just where walls of buildings should have given protection and the provision for under-deck car parking is costly and bleak.

The spine blocks

The spine block in stage 1 gives the impression of a continuous random pattern, but, like a clever wallpaper design, it in fact consists of fourteen 16ft bay units **C-K**



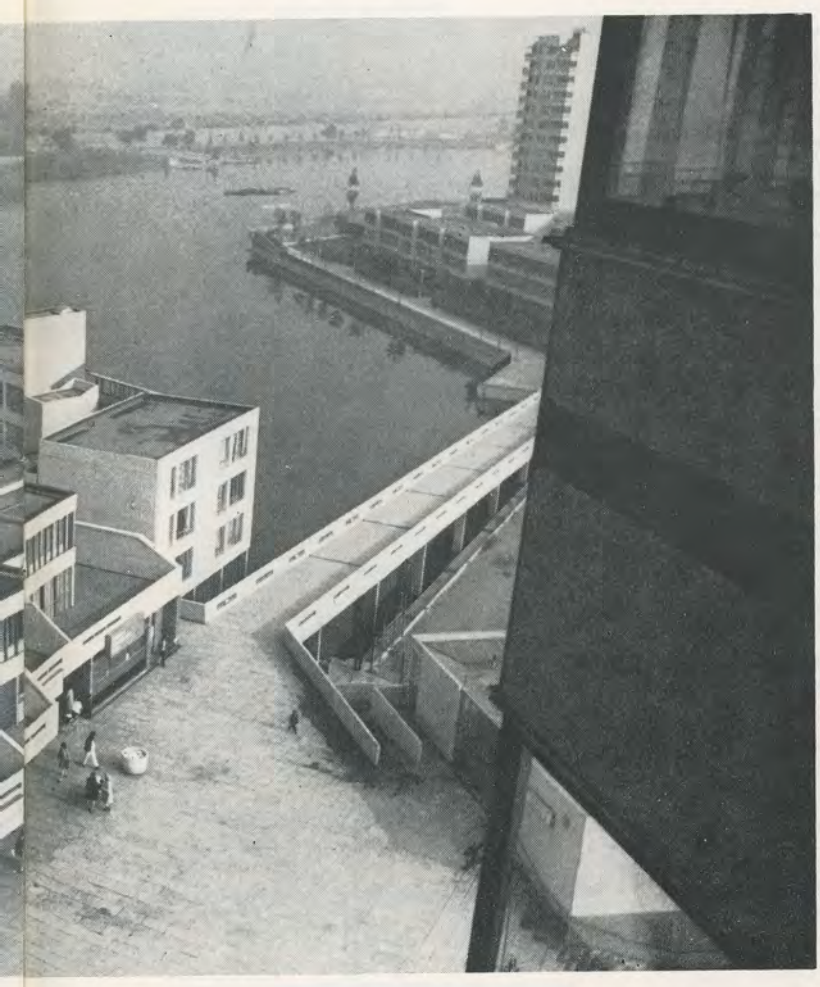
B *Typical floor plan of point block*



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which are reflected in the next 14 bays, so that the whole pattern only repeats through 28 bays. I find it impossible not to admire this invention. The craggy profile of the building is consistently sustained by indentations of plan and section and the angled projections of balconies, roof-lights, ramps and stairs **32**, **33**. The quality of concrete here, as throughout the scheme, with white aggregate and white cement, is surprisingly humane and pleasant.

It is like a giant kind of Lego set and seems to represent, visually, the idea of industrialised building as a kit of parts which can be combined in a variety of ways. It is ironical, then, to discover that it was not carried out in the Balency system. In preliminary studies at tender stage, Cubitts found that the design (which like the whole of stage I was conceived without any particular IB system in mind) was not economic in Balency. The numerous panel types would have been inordinately expensive in the special Balency steel moulds and were instead made in timber moulds and hung on an in situ concrete frame. The complexities of the design would also have posed difficulties for site management and for a labour force whose satisfaction is not in overcoming new difficulties each day but in seeing progress. In fact the construction of the spine block separately proved an advantage for it enabled a start to be made on stage I before the completion of the IB factory.

27 *Way into local centre across lake—also way in for cold north-east winds accentuated by tower*
28 *Grass court behind shops overlooked by old people's flats*
29 *Service road under centre*
30 *Point blocks elegantly express quality of Balency system*
31 *Ingenious detail: slot in balcony balustrade doubles, right, as kitchen ventilator*

Taking a cooler, analytical approach, one notes, as the architects have mentioned in their introductory account, that the spine block has a density little different from that of the three-storey low-rise housing. Each 14-bay unit accommodates 107 people. In a double row of terrace houses each side of a cul-de-sac, the particular dwelling mix in the spine block would give 4·5 people per unit frontage of dwelling, at a total of 126 people over the same length. If the spine blocks were arranged in rows, there would be more space between them but the angle of obstruction would be greater and the subjective effect more dense.

The reason for this similarity is that the spine blocks are not effectively two-sided streets. Half the bedrooms in the maisonettes have to have an outlook across the centre of the block, so they either stand opposite a complete void or overlook a single-storey dwelling. The spine block offers dwellings with more difficult access and smaller gardens at greater cost than the houses.

The spine block in stage II is a poor thing by comparison **L-N**, straightened out to bring down the cost, but still with balconies, which may not be achieved within the yardstick in stage III. Its redeeming feature is its magnificent relationship with the rolling grass hump of the linear park, against which it lies like a ship in the trough of a wave **19**.

Terrace houses

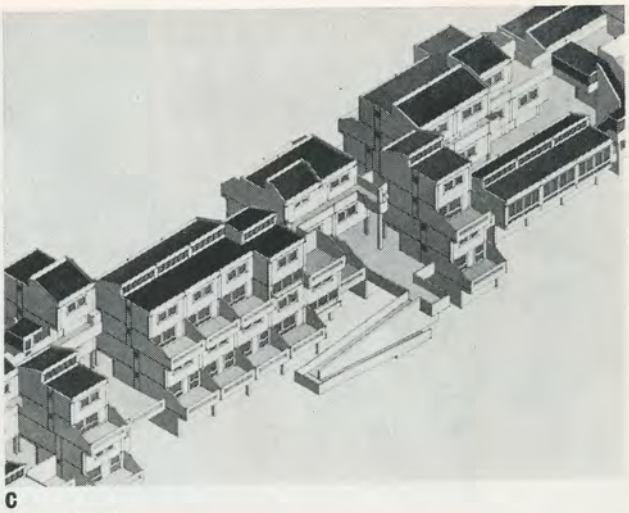
The terrace houses are 13ft frontage for four-person families and 15ft 9in to accommodate further bed spaces for five- and six-person families. These three-storey houses are really two-storey houses raised off the ground. They lack one of the usual merits of three-storey houses, which is that the entrance gives access to the stair and hence to all rooms without loss of privacy. With the access gallery at first floor level they suffer the disadvantage of the front door opening straight into the small dining area.

The larger houses have integral garages, the smaller houses, not related to culs-de-sac, instead overhang sheltered covered ways at ground level. In both stage I and II these make very pleasant cloisters of main pedestrian routes. But, in terms of yardstick penny pinching (which didn't apply to stage I and only partially applied to stage II) the duplication of covered routes at two levels, and the decision not to use such an obvious space for garaging, would seem very extravagant.

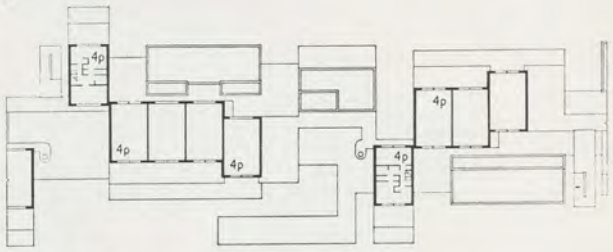
Under the other side of the house there is a covered open space, extending into the garden. Now that the by-law no longer applies this is potential space for expansion and garden rooms which some tenants have anticipated with personal arrangement of trellis work which the architecture is strong enough to accept **34**, **35**.

Like those of the spine blocks, the elevations of the terrace houses consist of two horizontal panels hung as cladding, rather than a single loadbearing panel with windows in it, like the towers. Floors are precast and internal cross walls give lateral rigidity to the party walls.

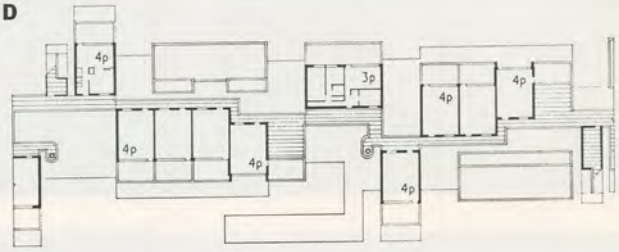
The strata of window and wall which characterise the spine blocks and terraces gives Thamesmead's housing a crispness and serenity **36** absent from the hole-in-the-wall window convention typical of the continental application of Balency. The windows themselves are peculiarly arranged. At bedroom level cedar mullions are set in a varied module, containing metal casements and enamelled infill panels. The effect is to distinguish the individual cell of space behind the concrete façade, implying that the dwellings are made with more transient and manipulable—and hence



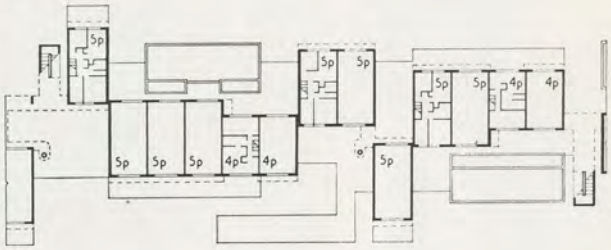
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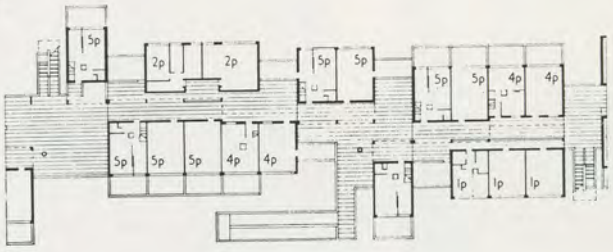
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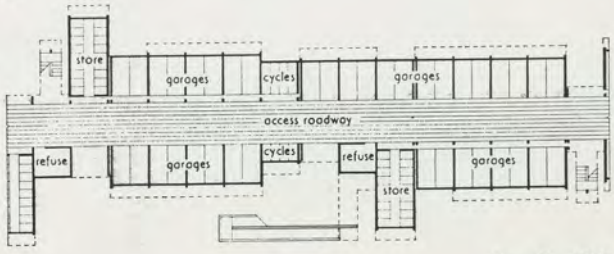
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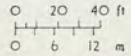
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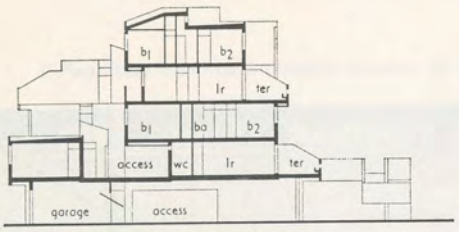


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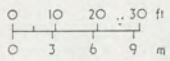


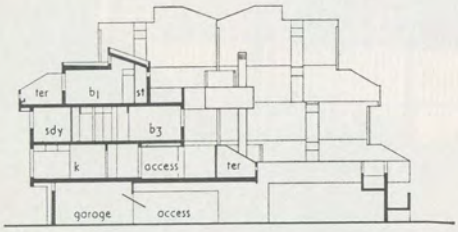
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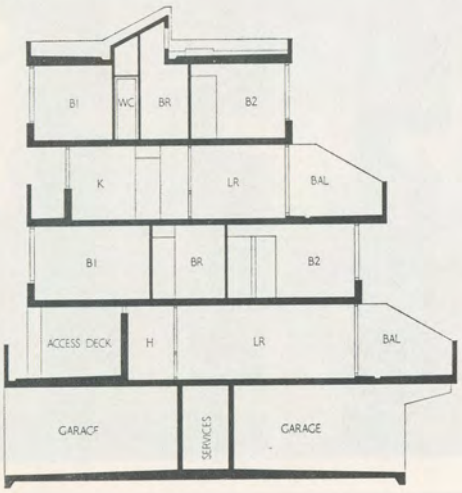


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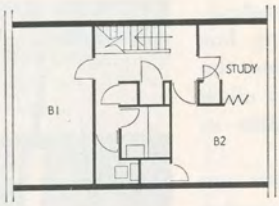


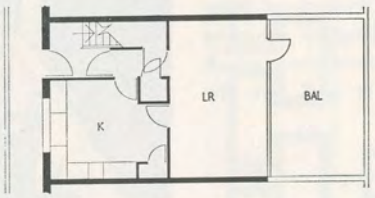


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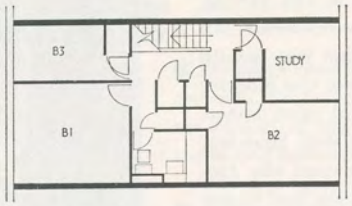


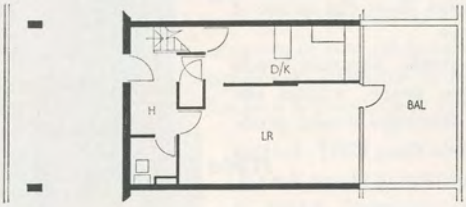
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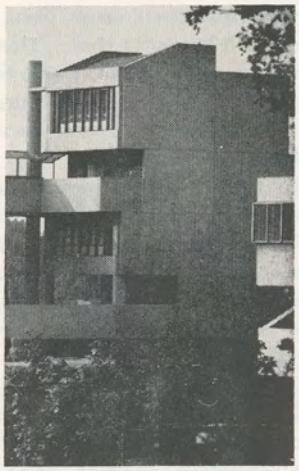


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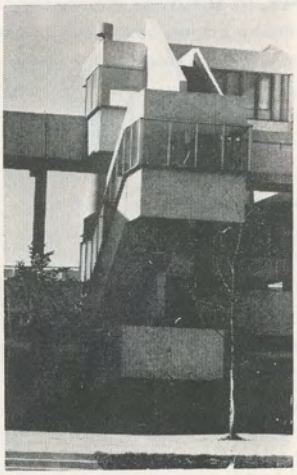




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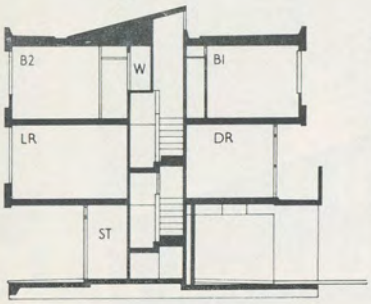


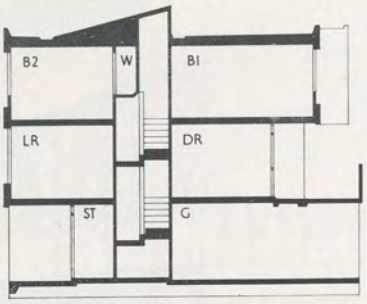
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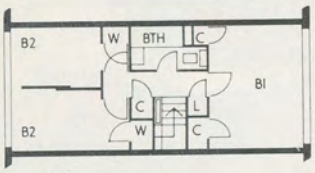


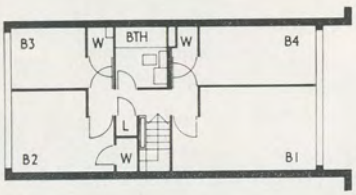
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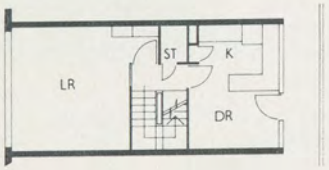
32, 33 *Craggy profile of spine, sustained by set-backs of plan and section, projections of balconies, rooflights, ramps and stairs*

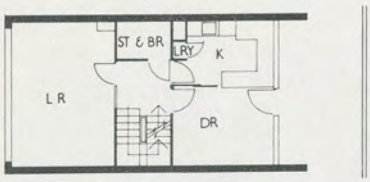






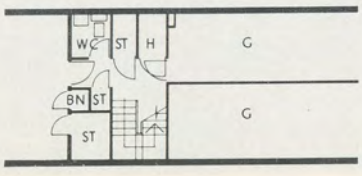








O



P

C-K *14-bay section of stage* I *linear block:*
C *Axonometric*
D *Fourth floor plan*
E *Third floor plan*
F *Second floor plan*
G *First floor plan*
H *Ground plan*
J, K *Typical cross sections; compare with (to double scale):*
L-N *Stage* II *linear block. The section shown is broken by link blocks containing three-person flats on three floors:*
L *Cross section (balconies face west)*
M *Upper (four person) maisonette*
N *Lower (five person) maisonette*
O, P *Typical three-storey house types in stage* I. *Stage* II *plans are similar, with modified roof section:*
O *Standard four-person house*
P *Six-person house; note wider bay dimension shared by five-person house*

more 'human'—elements than the systematic megastructure which supports them. This message is reassuring, but a deceit in so far as the system in fact rigidly determines the internal layout of the dwellings and will not readily allow the changes in plan and section which are possible in traditional construction.

At living room level normal centre-pivoted windows are introduced, underlining the superfluousness of the wooden mullion system above. Surely a universal system of fenestration could have been devised, consisting of essential elements, which might have survived the economics which have removed the mullions from stage II and left the elevations partially unresolved.

Density of the terrace houses and patterns of land use: a hypothetical alternative

It is interesting to find that the net density of these terrace houses, in the most elementary street layout is 154 ppa **Q**, a density which exceeds the average for Thamesmead. An objection to using solely terrace houses in this way would be the monotony of a large area. But if we take the Martin/March technique of arranging housing around open space, and apply this to the site area of stage II we find that low rise housing could be disposed in a band around the two primary school sites **R**, which would be less deep and therefore, in a simple sense, less monotonous than the areas of low rise housing in either stages I or II*.

To achieve this, accommodate access roads, hammer heads and additional car parking and open space, within the housing site area, the average house frontage would probably have to decrease though not to less than 12ft†. In this example the ratio of open space to housing is about 1 to 5. If the open space is proportionately increased, which it could be theoretically given the proportion in the master plan, the housing layout could open out and plot sizes increase in the way that was demonstrated in the previous article.

This arrangement reduces the depth of vehicular penetration of the housing area and brings all the housing into close contact with open space, so that people might feel they were at the edge of a park rather than in a tight urban area. More fundamentally, it dispenses with all that concrete construction which has been so ingeniously and expensively devised to hide cars under decks, raise dwellings off the ground and provide ramps, stairs and bridges to give access to them.

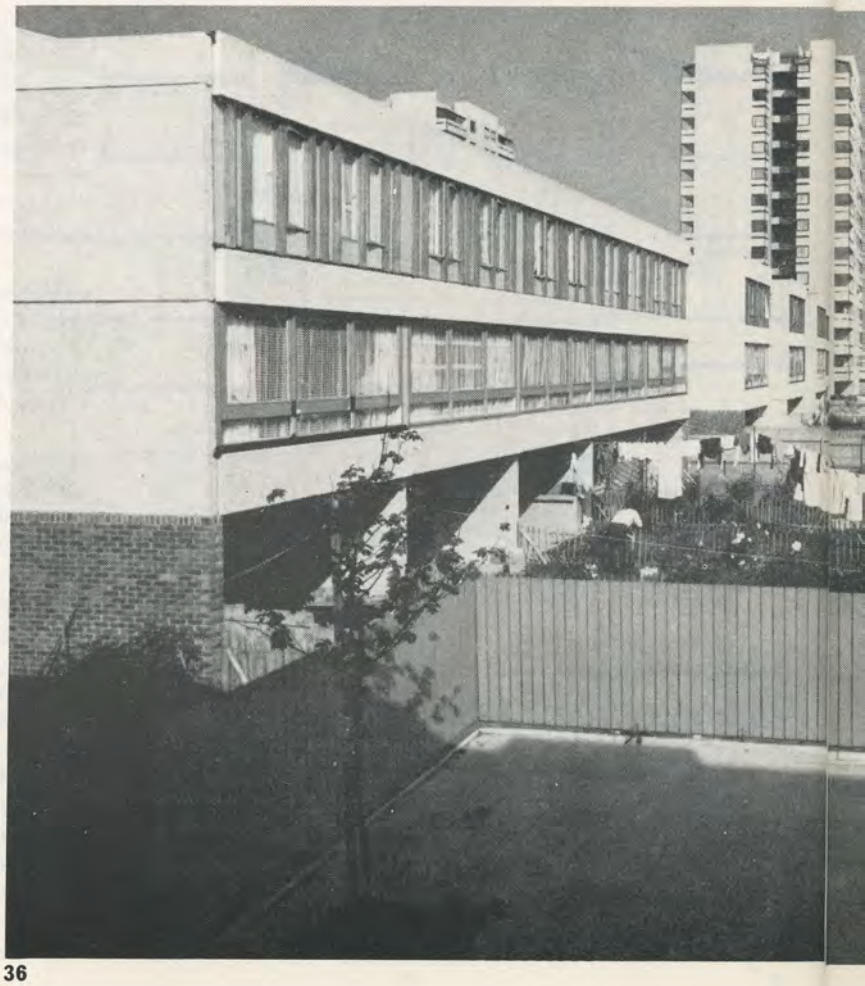
This appraisal has been critical where it has been concerned with alternative strategies rather than simply with the quality of what has been accomplished. It would be quite wrong to dismiss Thamesmead either because of this or because the 'style' of its architecture seems, in a very superficial sense, out of date. Thamesmead is industrialised building on a marsh; the achievement so far is that it has transcended the grim image which this combination evokes and invented a new environment. It is, in many ways, deeply considered, humane, even passionate, and this is largely what architecture is about.

* For further discussion of the properties of layout patterns see LIONEL MARCH 'An examination of layouts' *Built Environment* September 1972 p374 and RICHARD MACCORMAC 'Pollards Hill, The evolution of the design' *Architectural Design* October 1971 p617

† It would be fair to point out that five and six-person dwellings at 12ft frontage need ground floor living space which would have originally been prohibited by the Bexley by-law



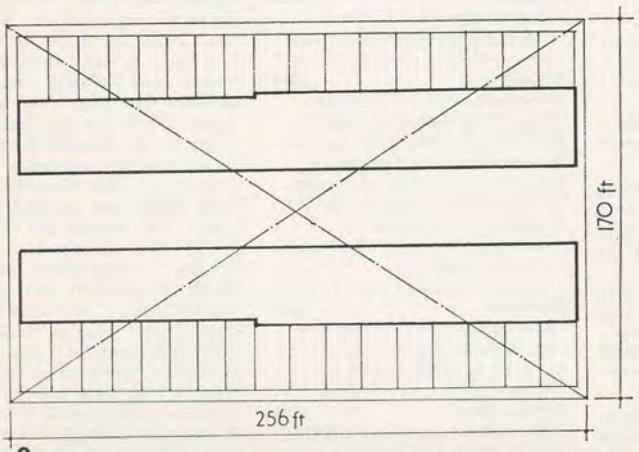
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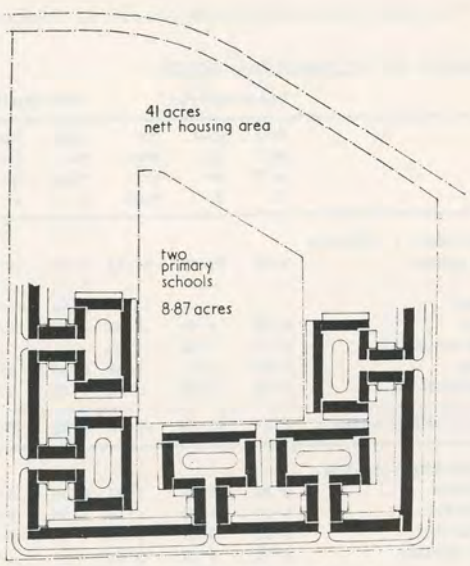
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34, 35 *Tenants' addition of trellis work across unused space under terrace houses*

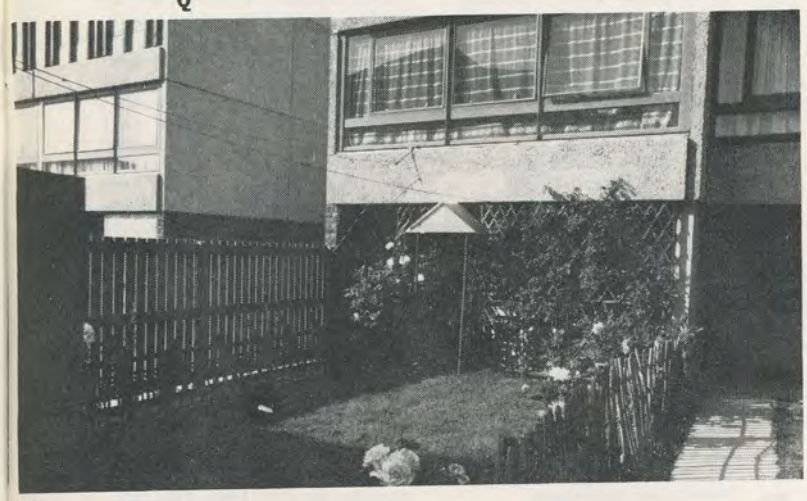
36 *Terrace house elevations: cedar mullions at bedroom level, metal windows for living rooms*



Q



R



35



Q *Diagram showing Thamesmead house types set out on one acre.* 18 *five-person and* 16 *four-person houses show a net density of* 154 *ppa. Additional car parking access roads and hammer-heads reduce this density to* 100 *ppa. Additional open space might reduce the frontage*
R *Diagrammatic layout for stage* II *by the author, based on principle explained* AJ 11.10.72 p828. *Both primary schools at centre of site, all housing in three storeys at* 12*ft frontage around the perimeter at* 100 *ppa. Vehicle access into housing reduced, open space close to most houses*

COSTS

No overall costs are yet available for Thamesmead stages I and II (and of course the enormous infrastructure costs of developing the site must be offset against the greatly enhanced value of the land). The GLC has, however, provided cost breakdowns of two typical dwellings (value cost contract, August 1966 price level). These we print below, with a comparative summary table of elemental costs per sq ft of net habitable floor area.

Note the large proportion of total cost in each case devoted to foundations, drainage and external works. This 'compendium' element includes pile and beam foundations to structure and drains; estate roads, sewers, district heating mains and pumping stations; access to garages and additional parking spaces. Also included for the two-person flat is a proportion of the paved open area at the foot of the block; for the four-person house, palisade fencing and gate to private garden, paved garden path and patio, covered walkway and a proportion of the courtyard and play area. These dwellings are not cheap; but a true comparison with elsewhere must include cost of land acquisition—which was:

SUMMARY: TWO-PERSON FLAT IN POINT BLOCK

Total floor area: 742 sq ft
Habitable floor area: 548 sq ft
Type of contract: value cost (August 1966 price level)
Tender date: August 1966
Work began: February 1967
Work continuing
Price of superstructure installation and finishes: £3746
Price of foundations, drainage and external works: £1251
Total £4977

COST BREAKDOWN

Preliminaries and insurances
3·5 per cent included with items

STRUCTURAL ELEMENTS

Balency system **£2075**
Precast internal and external walls with galvanised metal window and wood surround cast in. Precast staircase and in situ floors and roof including lift motor room and roof finish, soil stack and refuse chute: 82 sq yd floor area, £25·55 per sq yd (£30·58/m²)

Windows **£214**
Timber, private balcony screen and door, service balcony screen and door, lobby screen, with handrail and door and galvanised metal staircase screen. Including all painting, glazing and ironmongery: 142 sq ft, £1·51 per sq ft (£1·81/m²)

SUMMARY OF ELEMENTAL COSTS

	Two-person flat			Four-person house		
	Cost per sq ft £	Cost per m² £	Per cent of total	Cost per sq ft £	Cost per m² £	Per cent of total
STRUCTURAL ELEMENTS						
Balency system	3·79	40·78	55·26	2·63	28·30	44·58
Roof	—	—	—	0·42	4·52	7·12
Staircases	—	—	—	0·16	1·72	2·71
Windows	0·39	4·20	5·69	0·65	6·99	11·02
External doors	0·13	1·40	1·90	—	—	—
Partitions	0·04	0·43	0·58	0·07	0·75	1·19
Internal doors	0·24	2·58	3·50	0·32	3·44	5·42
Total of structural elements	4·59	49·39	66·92	4·25	45·73	72·04
FINISHES AND FITTINGS						
Wall finishes	0·08	0·86	1·17	0·09	0·97	1·53
Floor finishes	0·12	1·29	1·75	0·08	0·86	1·35
Ceiling finishes	0·03	0·32	0·44	0·03	0·32	0·51
Staircase finishes	0·02	0·22	0·29	—	—	—
Fittings	0·18	1·94	2·62	0·10	1·08	1·70
Total of finishes and fittings	0·43	4·63	6·27	0·30	3·23	5·09
SERVICES						
Sanitary fittings and services	0·15	1·61	2·19	0·31	3·34	5·25
Heating and ventilation services	0·88	9·47	12·83	0·59	6·35	10·00
Gas services	0·04	0·43	0·58	0·02	0·22	0·34
Electrical services	0·29	3·12	4·23	0·44	4·73	7·46
Special services	0·48	5·16	7·00	—	—	—
Total of services	1·84	19·80	26·83	1·36	14·63	23·05
Total	6·86	73·80	100·00	5·90	63·48	100·00
External works	2·28	24·53		3·09	33·25	

External doors £72
First floor main entrance doors. Dwelling entrance door and side panel including all painting, glazing and ironmongery: 35 sq ft, £2·06 per sq ft (£22·27/m²)

Partitions £21
Softwood framed stud partitioning and plasterboard: 12 sq yd, £1·75 per sq yd (£2·10/m²)

Internal doors £131
Pre-hung doors and frames (including to linen, store and heater cupboard) including all painting, glazing and ironmongery: 100 sq ft, £1·31 per sq ft (£14·67/m²)

FINISHES AND FITTINGS

Wall finishes £42
All wall finishes including lobby: 135 sq yd, £0·31 per sq yd (£0·41/m²)

Floor finishes £67
To flat and lobby dwelling floor: felt backed vinyl floor covering and vinyl preformed skirting: 60 sq yd, £1·11 per sq yd (£1·34/m²)

Ceiling finishes £14
All ceiling finishes including lobby: 64 sq yd, £0·22 per sq yd (£0·27/m²)

Staircase finishes £10
Mosaic landings with painted walls and sloping soffits

Fittings £97
Kitchen fittings, kitchen/living room screen and work top, all shelving and decorations

SERVICES

Sanitary appliances and services £82
including soil and vent

Heating and ventilation £482
including hot and cold services and dry riser installation

Gas installation £20

Electrical installation £158
including materials cast in panels and lightning conductor

Special services £261
Two lifts per block serving 12 floors

SUMMARY: STANDARD THREE-STOREY FOUR-PERSON HOUSE

Total floor area: 1263 sq ft
Habitable floor area: 915 sq ft
Type of contract: value cost (August 1966 place level)
Tender date: August 1966
Work began: February 1967
Price of superstructure, installation and finishes: £5401
Price of foundations, drainage and external works: £2828
Total: £8229

COST BREAKDOWN

Preliminaries and insurances
7·8 per cent included with items

STRUCTURAL ELEMENTS

Balency system £2411
Precast internal and external walls, floors, beams and access walkway £17·22 per sq yd (£20·59/m²)

Roof £380
Prefabricated timber construction with asphalt finish including clerestorey windows, glass and soffit installation: 45 sq yd, £8·44 per sq yd (£10·09/m²)

Staircase £146
Timber construction including painting and glazing to balustrade

Windows £599
Galvanised metal. Kitchen/diner screen, living room screen, and ground floor front and rear screens incorporating doors. Bedroom windows, metal in softwood frames with hardwood mullions including all painting, glazing and ironmongery: 398 sq ft, £1·55 per sq ft (£16·68/m²)

Partitions £61
Softwood framed stud partitioning and plasterboard: 35 sq yd, £1·74 per sq yd (£2·08/m²)

Internal doors £294
Prehung doors and frames including painting, glazing and ironmongery: 224 sq ft, £1·31 per sq ft (£14·1/m²).

FINISHES AND FITTINGS

Wall finishes £81
Decoration to concrete wall, dry lining and partitions.
Tiling to bath and basins: 264 sq yd, £0·31 per sq yd (£0·37/m²)

Floor finishes £74
Vinyl floor tiles and vinyl preformed skirtings: 85 sq yd, £0·87 per sq yd (£1·04/m²)

Ceiling finishes £26
Texture finish to concrete, and dry lining soffits: 92 sq yd, £0·36 per sq yd (£0·34/m²).

Fittings £89
Kitchen fittings including kitchen/diner work top. All shelving and decorations

SERVICES

Sanitary appliances and services £284
All fittings and plumbing, hot and cold, soil and vent and rwp

Heating and ventilating £536
Installation including heart unit and builder's work

Gas installation £21

Electrical installation £399
Harness system including builder's work

CONTRACTORS

Main contractor: Cubitts (Southern) Ltd. *Mosaic finishes:* Art Pavements & Decorations Ltd. *Screens to walkways and some buildings:* J. B. Abbott & Co (Plasterers) Ltd. *Mechanical services (linear housing):* F. J. Baynes & Co Ltd. *Aluminium roof flashings (linear housing):* E. W. Bathmaker. *Mastic pointing:* F. Brooks, Techni-seal (UK) Ltd and Shaw Seal. *Aluminium roof flashing (three-storey housing and linear housing):* Builders Iron & Zincwork Ltd. *Painting and decorating:* Alfred Bagnall & Sons Ltd. *Fencing and gates (three-storey housing):* Durafencing (London) Ltd. *Glazing:* Faulkner Green & Co Ltd, E. A. Heatherington Ltd and Shepherd Tobias Ltd. *Domestic flooring:* Floor Range Ltd. *Electrical installation:* Holliday Hall & Co Ltd. *Heating and soil and vent installation (linear housing):* G. N. Haden & Sons Ltd. *Texture finish to soffits (linear housing):* High Power Plastering Ltd. *Supply and fixing reinforcement:* Hy-ten Reinforcement Co Ltd. *Patent glazing (linear housing):* Heywood-Helliwell Ltd. *Plastering and dry lining:* T. J. Lott & Son Ltd. *Mechanical services and sanitary plumbing:* Matthew Hall Ltd. *Floor finishes:* Marley Tile Co Ltd and Northdown Contracts Ltd. *Screeding to walkways and balconies (linear housing):* Martin & Denis Ltd. *Wall tiling:* Parkinson (Wall tiling) Ltd. *Asphalt paving, roofing and tanking:* H. V. Smith & Co Ltd. *Landscaping:* Henry Shannon (Landscaping) and St Mary's (Contractors) Ltd. *Heating and ventilation, h and c water services (point blocks):* Taylor Engineering (Heating Services) Ltd. *Metal louvre windows to garages (linear and three-storey housing), metal windows (linear housing and point blocks):* Williams & Williams Ltd. *Piling:* Westpiling & Construction Co Ltd. *Balcony screens, grills and balustrades (point blocks):* Wallbays Ltd. *Patent glazing (central area and block 8):* S. Warner & Sons Ltd. *Road surfacing:* Wimpey Asphalt Ltd. *Suppliers: Aggregates:* British Dredging (London) Ltd, Hall & Co Ltd, Amalgamated Roadstone Corporation Ltd. *Facing bricks:* Moore, Scott & Co Ltd. *Partition blocks:* G. S. Warr-Wood Ltd. *Joinery:* J. C. Richards (Woodworkers) Ltd, Davies Bros. *Flush doors:* Manor Joinery. *Metalwork:* Metalcraft (Erith) Ltd. *Cement:* A. Turner & Sons Ltd, William Cory & Sons. *Ready mixed concrete:* Hall & Co Ltd, Readymix Ltd. *Plasterboard:* Hall & Co Ltd. *Timber and plywood:* John Lenanton & Co Ltd, Phoenix Timber Co Ltd. *Glazing frames:* Hitch Engineering Co Ltd, Haltrac Ltd. *Manhole covers and frames:* Thames Bank Iron Co Ltd. CI *pipe and fittings:* Stanton Staveley, Finch Froy Ltd, Thames Bank Iron Co Ltd. *Precast concrete manhole chambers:* Mono Concrete Ltd, Spun Concrete Ltd. *Spun concrete pipes and fittings:* Spun Concrete Ltd. *Aluminium flashings:* Aliform Ltd, Haltrac Ltd. *Garage doors:* P. C. Henderson Ltd. *Ironmongery:* Childs Constantine Ltd. *Kitchen units:* Remploy Ltd. *Kerbs and pavings:* Atlas Stone Co Ltd, ECC Quarries Ltd, Metropolitan Concrete Works Ltd. *Salt glazed pipes and fittings:* Doulton Vitrified Pipes, Hall & Co Ltd. *Concrete refuse chutes:* True Flue Ltd. *Reinforcing mesh:* BRC Engineering Co Ltd, Spencer Wire Co Ltd. *Reinforcing steel:* Square Grip Reinforcement Co Ltd, Rom River, GKN (South Wales) Ltd. Hy-ten Reinforcements Ltd, McCall & Co (Sheffield) Ltd. *Stainless steel rod:* Samuel Fox & Co Ltd, Macready's Metal Co Ltd, Arthur Lee & Sons Ltd. *Direct supply by* GLC: *Baths:* E. Curran (Engineering) Ltd. *Basins:* Armitage Ware Ltd, Shanks & Co Ltd. *Wc suites:* Shires Ltd. *Wc pans:* Shanks & Co Ltd. *Wcps:* Shires Ltd. *Electric fires:* Berry Magicoal Ltd. *Garage doors:* Morris Singer & Haskins Ltd. *Stainless steel sinks:* Fisher-Bendix Ltd.

PHOTOGRAPHS

All photos by Tim Street-Porter except: 9, 10, 15, 16, 20, 30, 36 by GLC photographic unit, Department of Architecture and Civic Design

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